

LinkedIn & IFC study of migration

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November 24, 2017

1 CONCEPTUAL FRAMEWORK

A city's productivity can be simply defined as the net result of contrasting forces $Productivity = Agglomeration(positive) - Congestion(negative)$

Where *Agglomeration* and *Congestion* are a function of various dimensions $= f(Skills, Amenities, Form, Access)$

- Dimensions
 - **Skills**, a city's aggregate stock of human capital
 - **Amenities** - *attracting skills* - job opportunities, housing values, cultural attractions
 - **Form**, the size and spatial configuration of a city (*density* vs. *sprawl*, *wider metropolitan areas*)
 - **Access**, a city's connectedness (or barriers) to other cities, both at home and abroad, through the transportation network
- This comparative analysis will focus on the first dimension **Skills**, with hints to the other ones

```
library(ggplot2) # install.packages("ggplot2")
library(dplyr)
library(knitr)
library(datasets) # initialize
library(knitr)
library(kableExtra)
library(stats)
library(tidyr)
library(stringr)
library(stats)

# getwd()
# migr <- read.csv("migration.csv", fileEncoding="UTF-8-BOM")
# demog <- read.csv("demographics.csv", fileEncoding="UTF-8-BOM")

load(here::here(".", "migr.Rdata"))
load(here::here(".", "demog.Rdata"))
```

2 DATA EXPLORATION BY COUNTRY

2.a Preliminary check on demog and migr

```
# names(demog)      #see all header (column) names
demog[1:5,]         # Indexing (1st to 5 th rows only)
```

##	NEW_MEM_ID	HIGHEST_DEGREE_OBTAINED	SENIORITY	EMPLOYER_INDUSTRY_SECTOR
## 1	1	doctor	Entry	Financial Services & Insurance
## 2	2	doctor	Partner	Architecture & Engineering

```
## 3      3      bachelor      CXO      Retail & Consumer Products
## 4      4      master      Entry      Technology - Hardware
## 5      5      master      Senior Government/Education/Non-profit
```

```
##      POSITION_FUNCTION
## 1 Information Technology
## 2 Business Development
## 3 Business Development
## 4 Information Technology
## 5 Education
```

```
str(demog)
```

```
## 'data.frame': 475316 obs. of 5 variables:
## $ NEW_MEM_ID : int 1 2 3 4 5 6 7 8 9 10 ...
## $ HIGHEST_DEGREE_OBTAINED : Factor w/ 4 levels "associate","bachelor",...: 3 3 2 4 4 2 2 2 4 3 ...
## $ SENIORITY : Factor w/ 11 levels "CX0","Director",...: 3 6 1 3 7 11 7 1 4 4 ...
## $ EMPLOYER_INDUSTRY_SECTOR: Factor w/ 14 levels "Aero/Auto/Transport",...: 3 2 10 12 4 14 7 7 4 4 ..
## $ POSITION_FUNCTION : Factor w/ 26 levels "Accounting","Administrative",...: 13 4 4 13 7 18 16
```

```
summary(demog) #see some summary statistics of each column
```

```
##      NEW_MEM_ID      HIGHEST_DEGREE_OBTAINED      SENIORITY
## Min. : 1      associate: 81072      Entry : 220187
## 1st Qu.: 118830      bachelor : 252952      Senior : 146986
## Median : 237658      doctor : 40502      Manager : 46279
## Mean : 237658      master : 100790      Training: 22047
## 3rd Qu.: 356487      Director: 21093
## Max. : 475316      VP : 7884
##      (Other) : 10840
##      EMPLOYER_INDUSTRY_SECTOR      POSITION_FUNCTION
## Government/Education/Non-profit: 85999      Engineering : 53792
## Technology - Software : 65725      Education : 41195
## Healthcare & Pharmaceutical : 63281      Sales : 39617
## Professional Services : 60623      Operations : 36840
## Financial Services & Insurance : 48297      Research : 31635
## Retail & Consumer Products : 36494      Information Technology: 31435
## (Other) : 114897      (Other) : 240802
```

```
# sapply(demog, class) # get class of all columns
```

```
# names(migr) #see all header (column) names
migr[1:5,] # Indexing (1st to 5 th rows only)
```

```
##      NEW_MEM_ID WEEK_BEGINNING SOURCE_COUNTRY      SOURCE_REGION
## 1      1      1/31/2016 United States      San Francisco Bay Area
## 2      2      9/18/2016 United States      Greater New York City Area
## 3      3      5/22/2016 United States      San Francisco Bay Area
## 4      4      7/24/2016 United States      Greater Detroit Area
## 5      5      1/24/2016 United States      Greater New York City Area
##      DESTINATION_COUNTRY      DESTINATION_REGION
## 1      United States      Greater Philadelphia Area
## 2      United Kingdom      London, United Kingdom
## 3      United States      Dallas/Fort Worth Area
## 4      United States      Greater Boston Area
## 5      United States      San Francisco Bay Area
```

```
str(migr)
```

```
## 'data.frame': 475316 obs. of 6 variables:
## $ NEW_MEM_ID : int 1 2 3 4 5 6 7 8 9 10 ...
## $ WEEK_BEGINNING : Factor w/ 53 levels "1/10/2016","1/17/2016",...: 5 51 34 43 3 45 52 31 46 46
## $ SOURCE_COUNTRY : Factor w/ 3 levels "Australia","United Kingdom",...: 3 3 3 3 3 3 3 3 3 3 ...
## $ SOURCE_REGION : Factor w/ 347 levels "Aberdeen, United Kingdom",...: 272 126 272 118 126 333 2
## $ DESTINATION_COUNTRY: Factor w/ 3 levels "Australia","United Kingdom",...: 3 2 3 3 3 3 3 3 3 3 ...
## $ DESTINATION_REGION : Factor w/ 282 levels "Abilene, Texas Area",...: 106 157 61 93 222 168 104 11 9
```

```
summary(migr) #see some summary statistics of each column
```

```
## NEW_MEM_ID WEEK_BEGINNING SOURCE_COUNTRY
## Min. : 1 7/31/2016: 11460 Australia : 8902
## 1st Qu.:118830 8/21/2016: 11433 United Kingdom: 36615
## Median :237658 8/14/2016: 11384 United States :429799
## Mean :237658 9/11/2016: 11184
## 3rd Qu.:356487 8/28/2016: 11182
## Max. :475316 8/7/2016 : 10916
## (Other) :407757
## SOURCE_REGION DESTINATION_COUNTRY
## Greater New York City Area: 37000 Australia : 8647
## Greater Los Angeles Area : 23070 United Kingdom: 36199
## San Francisco Bay Area : 22643 United States :430470
## Washington D.C. Metro Area: 19744
## Greater Chicago Area : 18521
## Greater Boston Area : 16898
## (Other) :337440
## DESTINATION_REGION
## Greater New York City Area: 38088
## San Francisco Bay Area : 36707
## Washington D.C. Metro Area: 25549
## Greater Los Angeles Area : 23362
## London, United Kingdom : 20136
## Greater Boston Area : 17909
## (Other) :313565
```

```
# sapply(migr, class) # get class of all columns
```

- **INSIGHTs:**

- Data contains 347 origin regions and only 282 destinations
- All Linkedin members (in data) have some tertiary education degree, ~ 50% are Entry level
- Linkedin members (in data) are distributed in 14 sectors

```
both <- left_join(demog,migr, by="NEW_MEM_ID")
```

Merge the 2 tables

2.b Frequencies and Proportions

I'm interested in studying members distribution across categorical variables according to origin country

Percentages and Proportions of *HIGHEST DEGREE* across countries of origin

```
# Single variable
country <- table(both$SOURCE_COUNTRY)
# Proportions for a single variable table
prop.table(country)

# Cross table by two variables
xcountry <- xtabs(~ HIGHEST_DEGREE_OBTAINED +SOURCE_COUNTRY, both)
# xcountry
addmargins(xcountry)

# Proportions in Cross Table
prop.table(xcountry)           # proportion to total
prop.table(xcountry, margin = 1) # proportion to row sum (DEGREE)
prop.table(xcountry, margin = 2) # proportion to column sum (ORIGIN COUNTRY)

# Stratified Table
## 3rd variable as stratified variable
xcountry2 <- xtabs(~ HIGHEST_DEGREE_OBTAINED +SENIORITY +SOURCE_COUNTRY, both)
xcountry2
## flat table
ftable(xcountry2)
```

Proportions of *HIGHEST DEGREE*/Seniority/Sector/Position against Country of origin with *Dplyr*

```
# Prop of members in each DEGREE to SUM of Country of origin
freq_OrigDegree <- both %>%
  group_by(both[,7],both[,2]) %>%
  summarise (n = n()) %>%
  mutate(freq = n / sum(n)) %>%
  mutate(rel.freq = paste0(round(100 * n/sum(n), 2), "%"))
```

`summarise()` has grouped output by 'both[, 7]'. You can override using the
`.groups` argument.

```
freq_OrigDegree #

## # A tibble: 12 x 5
## # Groups:   both[, 7] [3]
##   `both[, 7]` `both[, 2]`      n   freq rel.freq
##   <fct>      <fct>      <int> <dbl> <chr>
## 1 Australia  associate      255 0.0286 2.86%
## 2 Australia  bachelor     5632 0.633 63.27%
## 3 Australia  doctor        611 0.0686 6.86%
## 4 Australia  master       2404 0.270 27.01%
## 5 United Kingdom associate   1312 0.0358 3.58%
## 6 United Kingdom bachelor   22107 0.604 60.38%
## 7 United Kingdom doctor     3533 0.0965 9.65%
## 8 United Kingdom master     9663 0.264 26.39%
## 9 United States associate   79505 0.185 18.5%
## 10 United States bachelor  225213 0.524 52.4%
## 11 United States doctor    36358 0.0846 8.46%
## 12 United States master    88723 0.206 20.64%
```

```
freq_OrigSniority <- both %>%
  group_by(both[,7],both[,3]) %>%
  summarise (n = n()) %>%
  mutate(freq = n / sum(n)) %>%
  mutate(rel.freq = paste0(round(100 * n/sum(n), 2), "%"))
```

`summarise()` has grouped output by 'both[, 7]'. You can override using the
`.groups` argument.

```
freq_OrigSniority # % remarkably similar in terms of Seniority across countries of ORIGIN
```

```
## # A tibble: 31 x 5
## # Groups:   both[, 7] [3]
##   `both[, 7]` `both[, 3]`      n    freq rel.freq
##   <fct>      <fct>      <int>  <dbl> <chr>
## 1 Australia CX0          119 0.0134  1.34%
## 2 Australia Director      323 0.0363  3.63%
## 3 Australia Entry        3542 0.398   39.79%
## 4 Australia Manager     1150 0.129   12.92%
## 5 Australia Owner         80 0.00899  0.9%
## 6 Australia Partner       41 0.00461  0.46%
## 7 Australia Senior     3331 0.374   37.42%
## 8 Australia Training     154 0.0173  1.73%
## 9 Australia Unpaid        1 0.000112 0.01%
## 10 Australia VP          161 0.0181  1.81%
## # i 21 more rows
```

```
freq_OrigSector <- both %>%
  group_by(both[,7],both[,4]) %>%
  summarise (n = n()) %>%
  mutate(freq = n / sum(n)) %>%
  mutate(rel.freq = paste0(round(100 * n/sum(n), 2), "%"))
```

`summarise()` has grouped output by 'both[, 7]'. You can override using the
`.groups` argument.

```
freq_OrigSector # % remarkably similar in terms of sector across countries
```

```
## # A tibble: 42 x 5
## # Groups:   both[, 7] [3]
##   `both[, 7]` `both[, 4]`      n    freq rel.freq
##   <fct>      <fct>      <int>  <dbl> <chr>
## 1 Australia Aero/Auto/Transport      357 0.0401  4.01%
## 2 Australia Architecture & Engineering      805 0.0904  9.04%
## 3 Australia Financial Services & Insurance      926 0.104  10.4%
## 4 Australia Government/Education/Non-profit 1716 0.193  19.28%
## 5 Australia Healthcare & Pharmaceutical      723 0.0812  8.12%
## 6 Australia Manufacturing/Industrial      372 0.0418  4.18%
## 7 Australia Media & Entertainment      506 0.0568  5.68%
## 8 Australia Oil & Energy      251 0.0282  2.82%
## 9 Australia Professional Services     1481 0.166  16.64%
## 10 Australia Retail & Consumer Products      504 0.0566  5.66%
## # i 32 more rows
```

```
freq_OrigPosition <- both %>%
  group_by(both[,7],both[,5]) %>%
```

```
summarise (n = n()) %>%
mutate(freq = n / sum(n)) %>%
mutate(rel.freq = paste0(round(100 * n/sum(n), 2), "%"))
```

`summarise()` has grouped output by 'both[, 7]'. You can override using the
`.groups` argument.

```
freq_OrigPosition # % remarkably similar in terms of position across countries
```

```
## # A tibble: 78 x 5
## # Groups:   both[, 7] [3]
##   `both[, 7]` `both[, 5]`      n   freq rel.freq
##   <fct>      <fct>      <int> <dbl> <chr>
## 1 Australia Accounting    208 0.0234 2.34%
## 2 Australia Administrative 162 0.0182 1.82%
## 3 Australia Arts and Design 293 0.0329 3.29%
## 4 Australia Business Development 663 0.0745 7.45%
## 5 Australia Community and Social Services 439 0.0493 4.93%
## 6 Australia Consulting    247 0.0277 2.77%
## 7 Australia Education    521 0.0585 5.85%
## 8 Australia Engineering    830 0.0932 9.32%
## 9 Australia Entrepreneurship 112 0.0126 1.26%
## 10 Australia Finance      508 0.0571 5.71%
## # i 68 more rows
```

- **INSIGHTS:**

- US has a significantly higher # of Associates leaving (18% vs 3% and 4%)
- Proportions seem remarkably similar in terms of Seniority / Sector / Position across countries of ORIGIN
- At the country level there are no big differences... probably makes more sense looking at city or internally
- Wonder if this is a subset of real Linkedin members or if there are particular similarities in the English-speaking countries

Percentages and Proportions - Seniority/Sector/Position against Country of DESTINATION- with *Dplyr*

I do the same analysis but looking at DESTINATION * Very similar results as per Origin

```
# prop DEGREE by country orf destin
freq_DestDegree <- both %>%
  group_by(both[,9],both[,2]) %>%
  summarise (n = n()) %>%
  mutate(freq = n / sum(n)) %>%
  mutate(rel.freq = paste0(round(100 * n/sum(n), 2), "%"))
freq_DestDegree # US has a significantly higher # of Associates leaving (18% vs 3% and 4%)

# prop SENIORITY by country orf destin
freq_DestSniority <- both %>%
  group_by(both[,9],both[,3]) %>%
  summarise (n = n()) %>%
  mutate(freq = n / sum(n)) %>%
  mutate(rel.freq = paste0(round(100 * n/sum(n), 2), "%"))
freq_DestSniority # % remarkably similar in terms of Seniority across countries

# prop SECTOR by country orf destin
```

```

freq_DestSector <- both %>%
  group_by(both[,9],both[,4]) %>%
  summarise (n = n()) %>%
  mutate(freq = n / sum(n)) %>%
  mutate(rel.freq = paste0(round(100 * n/sum(n), 2), "%"))
freq_DestSector # % remarkably similar in terms of sector across countries

# prop POSITION by country orf destin
freq_DestPosition <- both %>%
  group_by(both[,9],both[,5]) %>%
  summarise (n = n()) %>%
  mutate(freq = n / sum(n)) %>%
  mutate(rel.freq = paste0(round(100 * n/sum(n), 2), "%"))
freq_DestPosition # % remarkably similar in terms of position across countries

# 3-Way Frequency Table
# mytable <- table(both[,4],both[,2], both[,3])
# ftable(mytable)

```

3 DATA EXPLORATION BY CITY

```

# create origin table
origin <- both %>% select(id=NEW_MEM_ID, degree=HIGHEST_DEGREE_OBTAINED, seniority=SENIORITY, sector=EM)

# create destination table
destination <- both %>% select(id=NEW_MEM_ID, degree=HIGHEST_DEGREE_OBTAINED, seniority=SENIORITY, sector=EM)

```

interim manipulation

3.1 Top 10 cities (US)

```

# Aggregate N flow (OUT) by City
library(knitr)
library(kableExtra)
aggreOrig <- origin %>%
  group_by(cityO) %>%
  summarize(NumOutflow= n()) %>%
  mutate(freq = NumOutflow / sum(NumOutflow)) %>%
  mutate(rel.freq = as.numeric(paste0(round(100 * NumOutflow/sum(NumOutflow), 3)))) %>%
  arrange(desc(NumOutflow))

# Top 10 ORIGIN cities
aggreOrig_short <- aggreOrig[1:10,]
kable(aggreOrig_short, #format = "html",
      caption = "Ranking of cities by Origin") %>% kable_styling(bootstrap_options = c("striped", "hover"))

```

Table 1: Ranking of cities by Origin

cityO	NumOutflow	freq	rel.freq
Greater New York City Area	37000	0.0778430	7.784
Greater Los Angeles Area	23070	0.0485361	4.854

San Francisco Bay Area	22643	0.0476378	4.764
Washington D.C. Metro Area	19744	0.0415387	4.154
Greater Chicago Area	18521	0.0389657	3.897
Greater Boston Area	16898	0.0355511	3.555
Dallas/Fort Worth Area	11592	0.0243880	2.439
Greater Philadelphia Area	11407	0.0239988	2.400
Greater Atlanta Area	11092	0.0233361	2.334
London, United Kingdom	10249	0.0215625	2.156

```
# create destination table
destination <- both %>% select(id=NEW_MEM_ID, degree=HIGHEST_DEGREE_OBTAINED, seniority=SENIORITY, sect

# Aggregate N flow (IN) by City
library(knitr)
library(kableExtra)
aggreDest <- destination %>%
  group_by(cityD) %>%
  summarize(NumInflow= n()) %>%
  mutate(freq = NumInflow / sum(NumInflow)) %>%
  mutate(rel.freq = as.numeric(paste0(round(100 * NumInflow/sum(NumInflow), 3)))) %>%
  arrange(desc(NumInflow))

# Top 10 DESTINATION cities
aggreDest_short <- aggreDest[1:10,]
kable(aggreDest_short, #format = "html",
      caption = "Ranking of cities by popular destination") %>% kable_styling(bootstrap_options = c("st
```

Table 2: Ranking of cities by popular destination

cityD	NumInflow	freq	rel.freq
Greater New York City Area	38088	0.0801320	8.013
San Francisco Bay Area	36707	0.0772265	7.723
Washington D.C. Metro Area	25549	0.0537516	5.375
Greater Los Angeles Area	23362	0.0491505	4.915
London, United Kingdom	20136	0.0423634	4.236
Greater Boston Area	17909	0.0376781	3.768
Greater Chicago Area	16944	0.0356479	3.565
Dallas/Fort Worth Area	16157	0.0339921	3.399
Greater Seattle Area	15553	0.0327214	3.272
Greater Atlanta Area	14242	0.0299632	2.996

- **INSIGHTS:**

- Intristingly, London is # 5 Origin but # 10 Destination
- 9/10 top DESTINATION are the same as top ORIGIN which suggest there is mobility, but not necessarily the top destination are places where people stay
- this can be explained by the American way of moving to and from the city of college
- *those where all american !!!*

3.2 Top 10 cities (UK)

```
# Aggregate N flow (OUT) by City
library(knitr)
library(kableExtra)

# subset origin
originUK <- subset(origin, countryO == "United Kingdom")

aggreOrigUK <- originUK %>%
  group_by(cityO) %>%
  summarize(NumOutflow= n()) %>%
  mutate(freq = NumOutflow / sum(NumOutflow)) %>%
  mutate(rel.freq = as.numeric(paste0(round(100 * NumOutflow/sum(NumOutflow), 3)))) %>%
  arrange(desc(NumOutflow)) #>% left_join(origin, by="cityO")

# Top 10 ORIGIN cities
aggreOrig_shortUK <- aggreOrigUK[1:10,]
kable(aggreOrig_shortUK, #format = "html",
      caption = "Ranking of cities by Origin") %>% kable_styling(bootstrap_options = c("striped", "hover"))
```

Table 3: Ranking of cities by Origin

cityO	NumOutflow	freq	rel.freq
London, United Kingdom	10249	0.2799126	27.991
Manchester, United Kingdom	1170	0.0319541	3.195
Reading, United Kingdom	1002	0.0273658	2.737
Twickenham, United Kingdom	990	0.0270381	2.704
Oxford, United Kingdom	978	0.0267104	2.671
Birmingham, United Kingdom	836	0.0228322	2.283
Guildford, United Kingdom	828	0.0226137	2.261
Kingston upon Thames, United Kingdom	788	0.0215212	2.152
Coventry, United Kingdom	695	0.0189813	1.898
Cambridge, United Kingdom	668	0.0182439	1.824

```
# =====#

# subset destination
destinationUK <- subset(destination, countryD == "United Kingdom")

# Aggregate N flow (IN) by City
library(knitr)
library(kableExtra)
aggreDestUK <- destinationUK %>%
  group_by(cityD) %>%
  summarize(NumInflow= n()) %>%
  mutate(freq = NumInflow / sum(NumInflow)) %>%
  mutate(rel.freq = as.numeric(paste0(round(100 * NumInflow/sum(NumInflow), 3)))) %>%
  arrange(desc(NumInflow))

# Top 10 DESTINATION cities
```

```

aggreDest_shortUK <- aggreDestUK[1:10,]
kable(aggreDest_shortUK, #format = "html",
      caption = "Ranking of cities by popular destination") %>% kable_styling(bootstrap_options = c("st

```

Table 4: Ranking of cities by popular destination

cityD	NumInflow	freq	rel.freq
London, United Kingdom	20136	0.5562585	55.626
Manchester, United Kingdom	1859	0.0513550	5.136
Birmingham, United Kingdom	949	0.0262162	2.622
Edinburgh, United Kingdom	803	0.0221829	2.218
Bristol, United Kingdom	754	0.0208293	2.083
Reading, United Kingdom	754	0.0208293	2.083
Cambridge, United Kingdom	719	0.0198624	1.986
Leeds, United Kingdom	695	0.0191994	1.920
Glasgow, United Kingdom	682	0.0188403	1.884
Oxford, United Kingdom	554	0.0153043	1.530

- **INSIGHTs:**
 - Contrary to widespread mobility in US, London is origin of 28% of migrants and the Destination for 55% of them

3.3 Top 10 cities (Austr)

```

# Aggregate N flow (OUT) by City
library(knitr)
library(kableExtra)

# subset origin
originAustr <- subset(origin, country0 == "Australia")

aggreOrigAustr <- originAustr %>%
  group_by(city0) %>%
  summarize(NumOutflow= n()) %>%
  mutate(freq = NumOutflow / sum(NumOutflow)) %>%
  mutate(rel.freq = as.numeric(paste0(round(100 * NumOutflow/sum(NumOutflow), 3)))) %>%
  arrange(desc(NumOutflow)) #>% left_join(origin, by="city0")

# Top 10 ORIGIN cities
aggreOrig_shortAustr <- aggreOrigAustr[1:10,]
kable(aggreOrig_shortAustr, #format = "html",
      caption = "Ranking of cities by Origin") %>% kable_styling(bootstrap_options = c("striped", "hov

```

Table 5: Ranking of cities by Origin

cityO	NumOutflow	freq	rel.freq
Sydney Area, Australia	3209	0.3604808	36.048
Brisbane Area, Australia	1995	0.2241069	22.411
Perth Area, Australia	1014	0.1139070	11.391
Adelaide Area, Australia	611	0.0686363	6.864
Canberra Area, Australia	556	0.0624579	6.246

Queensland, Australia	468	0.0525725	5.257
New South Wales, Australia	357	0.0401033	4.010
Newcastle Area, Australia	216	0.0242642	2.426
Western Australia, Australia	103	0.0115704	1.157
Toowoomba Area, Australia	74	0.0083127	0.831

```
# =====#

# subset destination
destinationAustr <- subset(destination, countryD == "Australia")

# Aggregate N flow (IN) by City
library(knitr)
library(kableExtra)
aggreDestAustr <- destinationAustr %>%
  group_by(cityD) %>%
  summarize(NumInflow= n()) %>%
  mutate(freq = NumInflow / sum(NumInflow)) %>%
  mutate(rel.freq = as.numeric(paste0(round(100 * NumInflow/sum(NumInflow), 3)))) %>%
  arrange(desc(NumInflow))

# Top 10 DESTINATION cities
aggreDest_shortAustr <- aggreDestAustr[1:10,]
kable(aggreDest_shortAustr, #format = "html",
      caption = "Ranking of cities by popular destination") %>% kable_styling(bootstrap_options = c("st
```

Table 6: Ranking of cities by popular destination

cityD	NumInflow	freq	rel.freq
Sydney Area, Australia	3885	0.4492888	44.929
Brisbane Area, Australia	2046	0.2366139	23.661
Canberra Area, Australia	674	0.0779461	7.795
Perth Area, Australia	532	0.0615242	6.152
Queensland, Australia	452	0.0522725	5.227
New South Wales, Australia	393	0.0454493	4.545
Adelaide Area, Australia	306	0.0353880	3.539
Newcastle Area, Australia	180	0.0208165	2.082
Western Australia, Australia	110	0.0127212	1.272
Toowoomba Area, Australia	69	0.0079796	0.798

- **INSIGHTs:**

- Similar to UK, in Australia Sydney is the origin of 36% of migrants and the Destination for 50% of them

3.4 Plots of top cities

```
# Adding some variables for plots
aggreByCity <- full_join(aggreDest, aggreOrig, by = c("cityD" = "cityO"))
aggreByCity[c("NumInflow", "NumOutflow")] [is.na(aggreByCity[c("NumInflow", "NumOutflow")])] <- 0
```

```

aggreByCity <- aggreByCity %>%
  select(-freq.x, -rel.freq.x, -freq.y, -rel.freq.y) %>% # get rid of meaningless
  mutate (NetFlow= NumInflow -NumOutflow) %>% # Net
  mutate (NegOutFlow= -(NumOutflow)) %>% # neg sign
  mutate (Sign = ifelse(NetFlow > 0, "Positive", "Negative")) %>%
  mutate (colour= ifelse(NetFlow > 0, "positive", "negative")) %>% mutate (city_copy = cityD) %>%
  separate(city_copy, into = c("city_only", "metro area"), sep = ",")

summary(aggreByCity)

```

JOIN the AGGREGATE by CITY to have Net flows in the same table (by City)

```

# ===== INTERIM ===== #
names(aggreByCity)[1]<-"city"
names(migr)[4]<-"city"
# city_country <-full_join(aggreByCity, migr, by = c("city", "SOURCE_REGION")) %>% # #select(city,coun

city_country <-full_join(aggreByCity, migr, by = "city") %>%
  select(city, country=DESTINATION_COUNTRY) %>%
  distinct () %>% # need the dup flag
  mutate(dupli = ifelse((city == "Greater New York City Area" | city == "San Francisco Bay Area" | city == "W
city == "Greater Los Angeles Area" | city == "Greater Boston Area" | city == "Greater Chicago Area" |
city == "Dallas/Fort Worth Area" | city == "Greater Seattle Area" | city == "San Francisco Bay Area") & co

  mutate(dupli2 = ifelse((city == "London, United Kingdom") & country != "United kindom", "dup", "")) %>%
  mutate(dupli3 = ifelse((city == "Sydney Area, Australia") & country != "Australia", "dup", "")) %>% # ne
  mutate(dupli4 = ifelse((city == "Perth Area, Australia") & country != "Australia", "dup", "")) %>% # ne
  mutate(dupli5 = ifelse((city == "Miami/Fort Lauderdale Area") & country != "United States", "dup", "")) %
  mutate(dupli6 = ifelse((city == "Brisbane Area, Australia") & country != "Australia", "dup", ""))

city_country <- subset(city_country, city_country$dupli != "dup" & city_country$dupli2 != "dup" & city_coun

newRow <- data.frame(city = "London, United Kingdom", country = "United Kingdom" )

city_country <- rbind(city_country, newRow)
#city_country <- rbind(city_country, c("London, United Kingdom", "United kindom"))

names(aggreByCity)[1]<-"cityD"
# ===== #

```

interim

Major 30 (World - mostly US) cities TO people are migrating

```

names(aggreByCity)[1]<-"city"

# 1) In Flow in Top DESTINATION

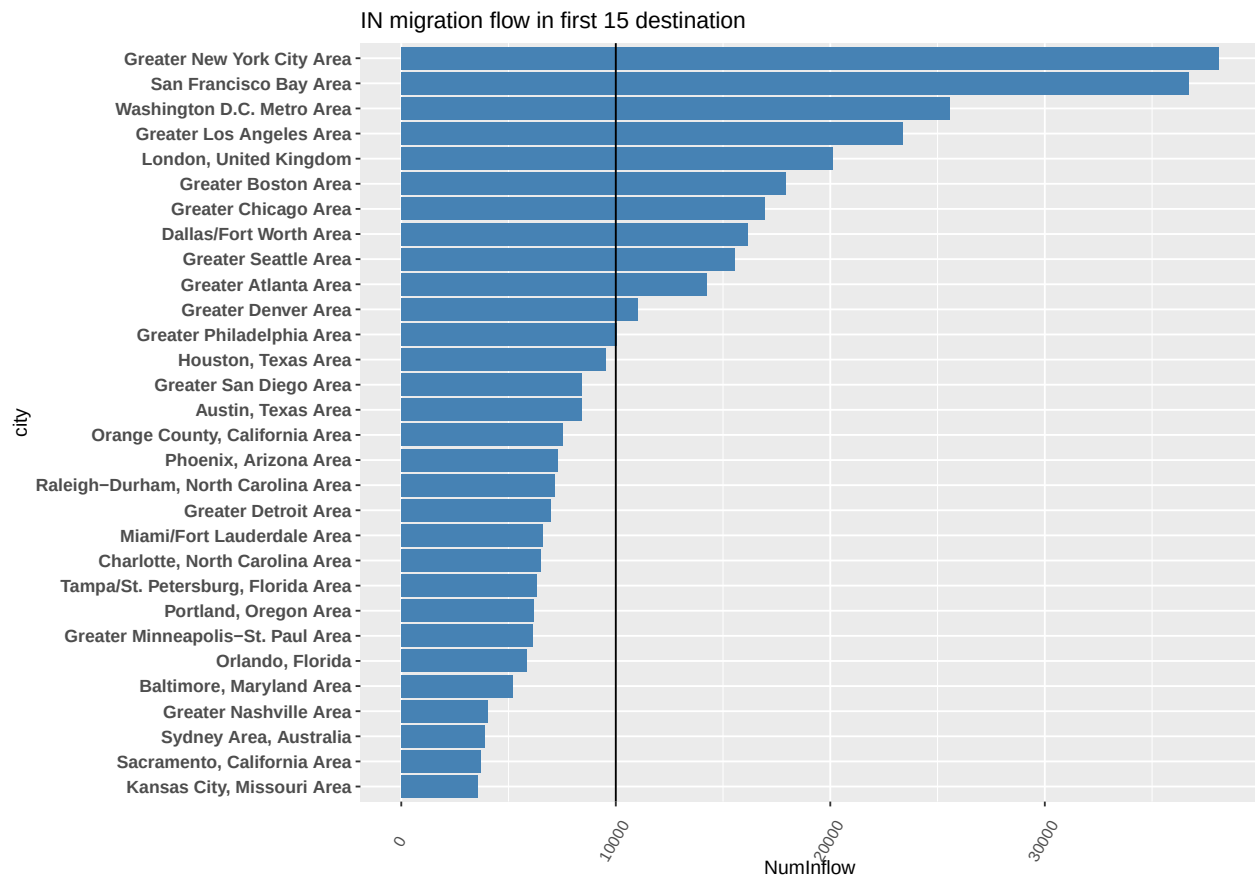
```

```
Top_in <- aggregateByCity %>% top_n(30, NumInflow)
Top_in$city = with(Top_in, reorder(city, NumInflow)) # reorder Levels by Var

Top_in <- ggplot(data = Top_in, aes(city, NumInflow)) +
  geom_bar(stat = "identity", position="identity", fill = "steelblue") +
  geom_hline(yintercept=10000, color = "black", size=0.5) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10, face="bold"))
labs(title="IN migration flow in first 15 destination")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

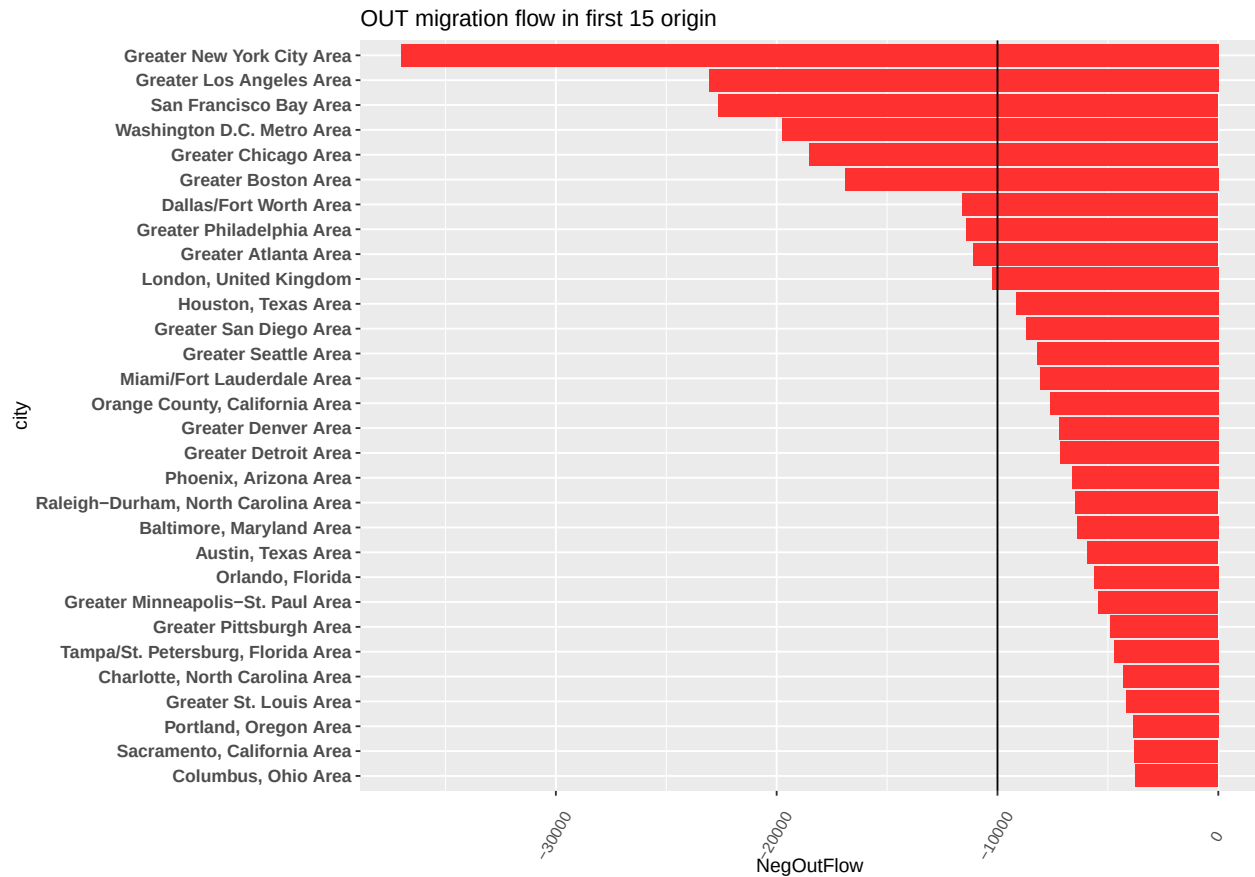
```
Top_in + coord_flip()
```



```
# 2) Out Flow in Top ORIGIN
Top_out <- aggregateByCity %>% top_n(30, NumOutflow)
Top_out$city = with(Top_out, reorder(city, NumOutflow)) # reorder Levels by Var

Top_out <- ggplot(data = Top_out, aes(city, NegOutFlow)) +
  geom_bar(stat = "identity", position="identity", fill = "firebrick1") +
  geom_hline(yintercept=-10000, color = "black", size=0.5) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10, face="bold"))
labs(title="OUT migration flow in first 15 origin")
```

```
Top_out + coord_flip()
```

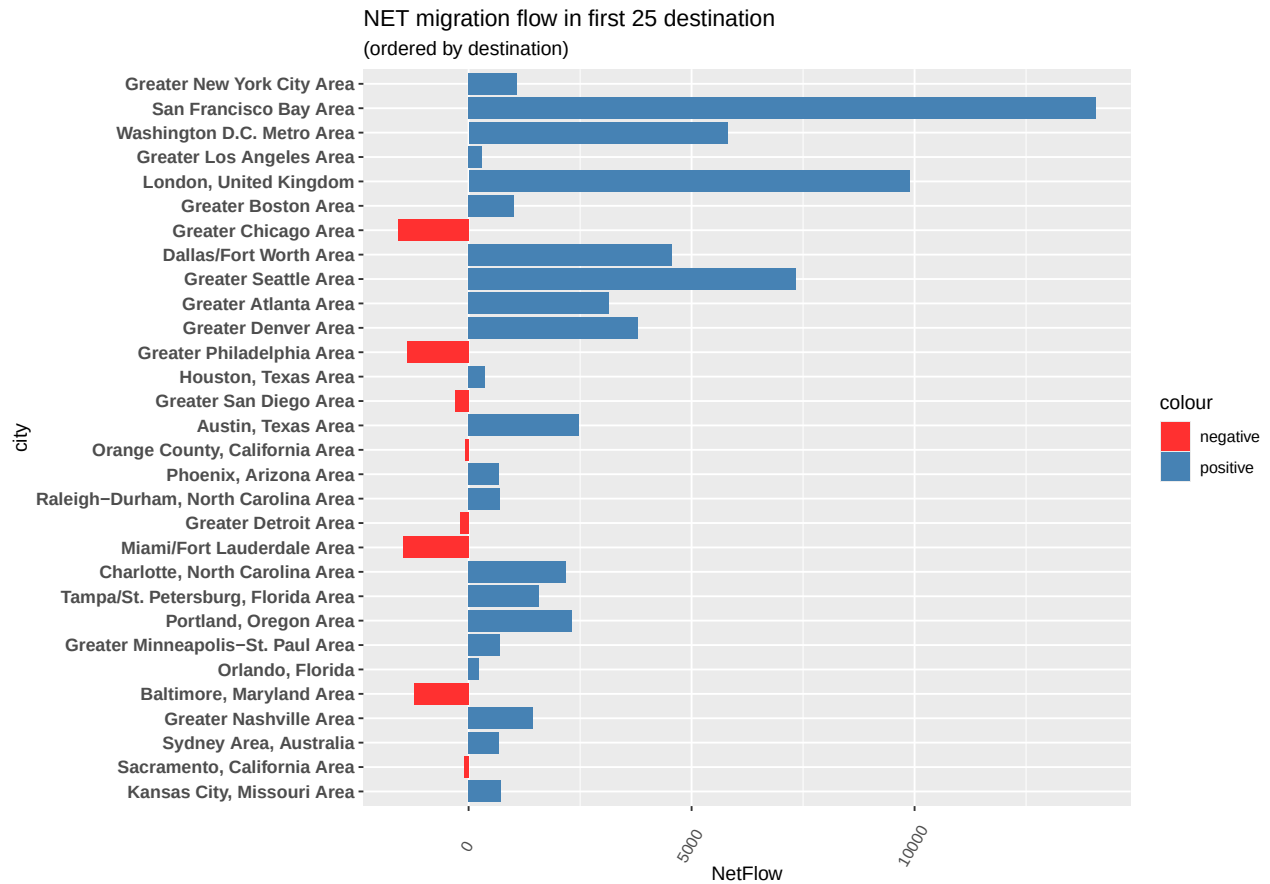


```
# plot side by side
# library(gridExtra)
# grid.arrange(in_flip, out_flip, ncol=2)

# 3.a) NET flow in Top DESTINATION
Top_net <- aggreByCity %>% top_n(30, NumInflow)
Top_net$city = with(Top_net, reorder(city, NumInflow)) # reorder Levels by Var

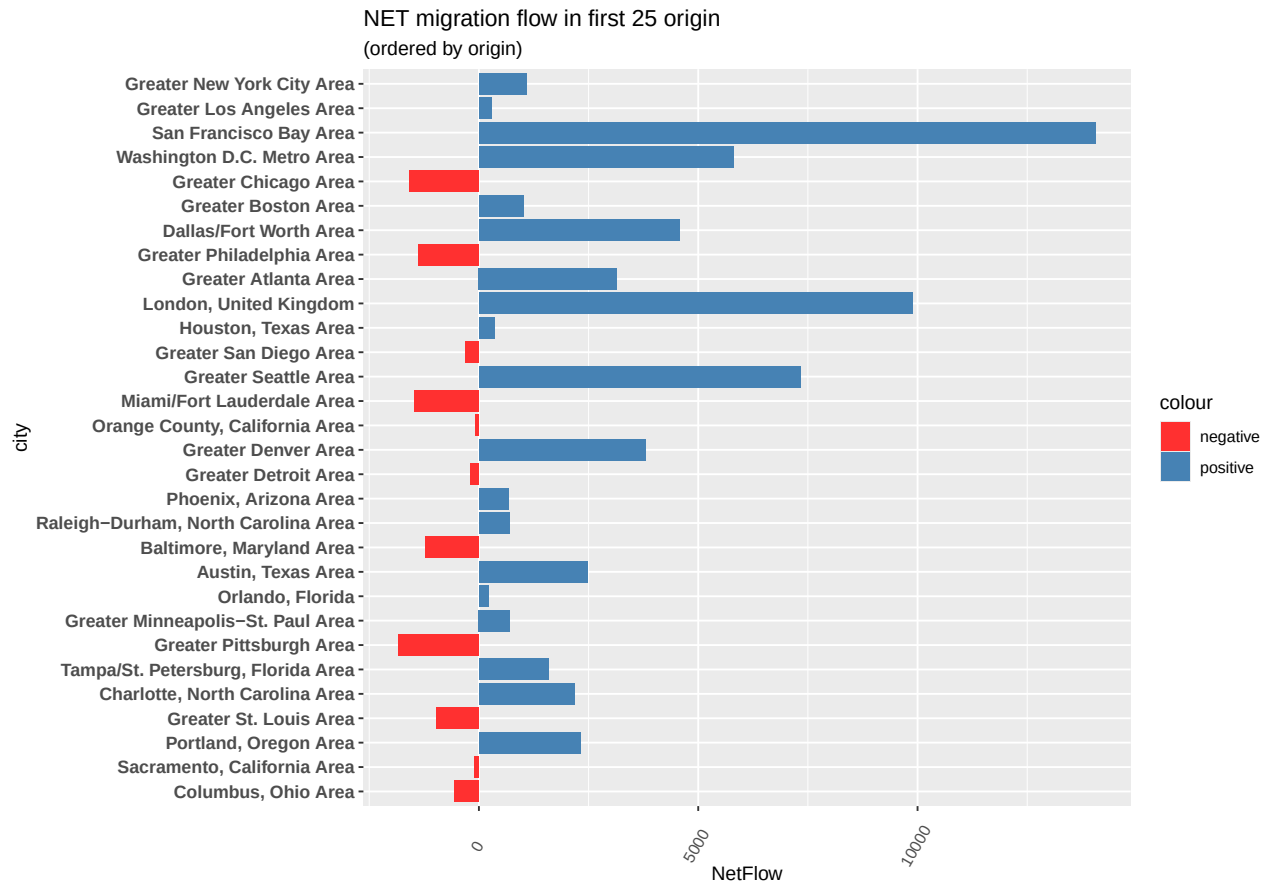
Top_net <- ggplot(data = Top_net, aes(city, NetFlow)) +
  geom_bar(stat = "identity", position="identity", aes(fill = colour)) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10, face="bold"),
  scale_fill_manual(values=c(positive="steelblue", negative="firebrick1")) +
  labs(title="NET migration flow in first 25 destination", subtitle="(ordered by destination)")

Top_net + coord_flip()
```



```
# 3.b) NET flow in Top ORIGIN
Top_net <- aggreByCity %>% top_n(30, NumOutflow)
Top_net$city = with(Top_net, reorder(city, NumOutflow)) # reorder Levels by Var
```

```
Top_net <- ggplot(data = Top_net, aes(city, NetFlow)) +
  geom_bar(stat = "identity", position="identity", aes(fill = colour)) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10, face="bold")) +
  scale_fill_manual(values=c(positive="steelblue", negative="firebrick1")) +
  labs(title="NET migration flow in first 25 origin", subtitle="(ordered by origin)")
Top_net + coord_flip()
```



```
# all american
# lots of leaving in 2016 - especially in NY

# add country to aggreByCity
aggreByCity2 <- left_join(aggreByCity,city_country,by = "city")

# x[c("a", "b")][is.na(x[c("a", "b")])] <- 0
aggreByCity2[c("NumInflow", "NumOutflow")][is.na(aggreByCity2[c("NumInflow", "NumOutflow")])] <- 0
```

Major 20 (UK) cities TO / FROM / NET migration

```
# 1) In Flow in Top DESTINATION

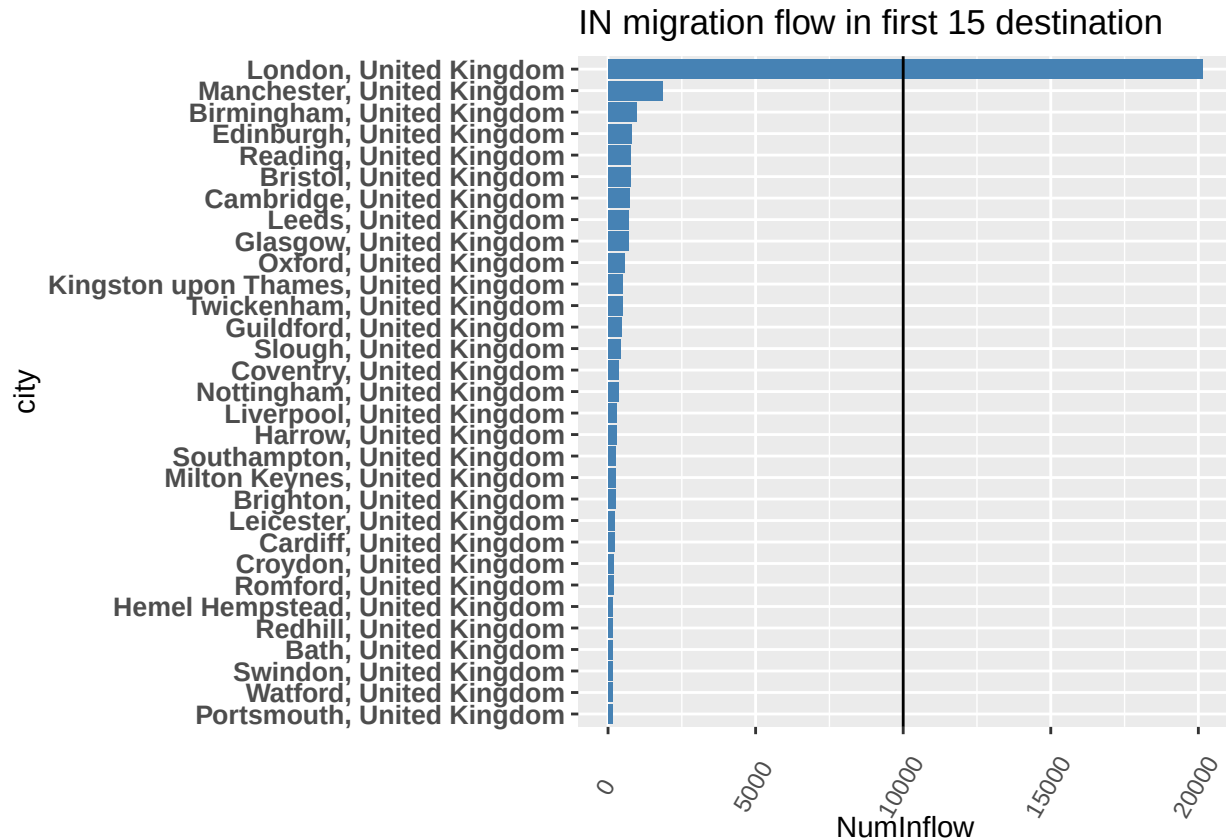
aggreByCityUK <- aggreByCity2 %>%
  filter (country == "United Kingdom")

Top_inUK <- aggreByCityUK %>%
  top_n(30, NumInflow)

Top_inUK$city = with(Top_inUK, reorder(city,NumInflow)) # reorder Levels by Var

Top_inUK <- ggplot(data = Top_inUK,aes(city, NumInflow)) +
  geom_bar(stat = "identity", position="identity", fill = "steelblue") +
  geom_hline(yintercept=10000, color = "black", size=0.5) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10,face="bold"),
  labs(title="IN migration flow in first 15 destination")
```

```
Top_inUK + coord_flip()
```



```
# 2) Out Flow in Top ORIGIN
```

```
Top_outUK <- aggreByCityUK %>%
```

```
  filter (country == "United Kingdom") %>% # filter by country = UK
```

```
  top_n(30, NumOutflow)
```

```
  Top_outUK$city = with(Top_outUK, reorder(city, NumOutflow)) # reorder Levels by Var
```

```
Top_outUK <- ggplot(data = Top_outUK, aes(city, NegOutFlow)) +
```

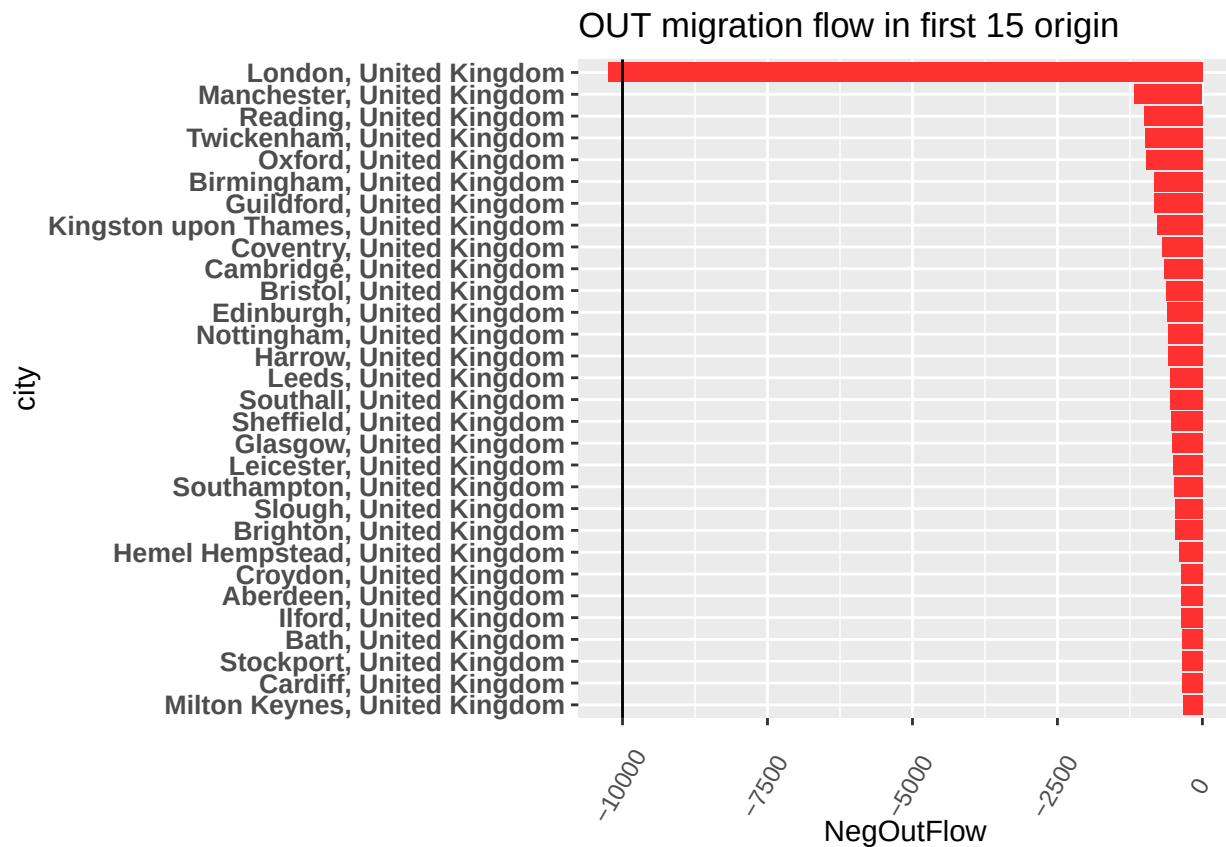
```
  geom_bar(stat = "identity", position="identity", fill = "firebrick1") +
```

```
  geom_hline(yintercept=-10000, color = "black", size=0.5) +
```

```
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10, face="bold"))
```

```
  labs(title="OUT migration flow in first 15 origin")
```

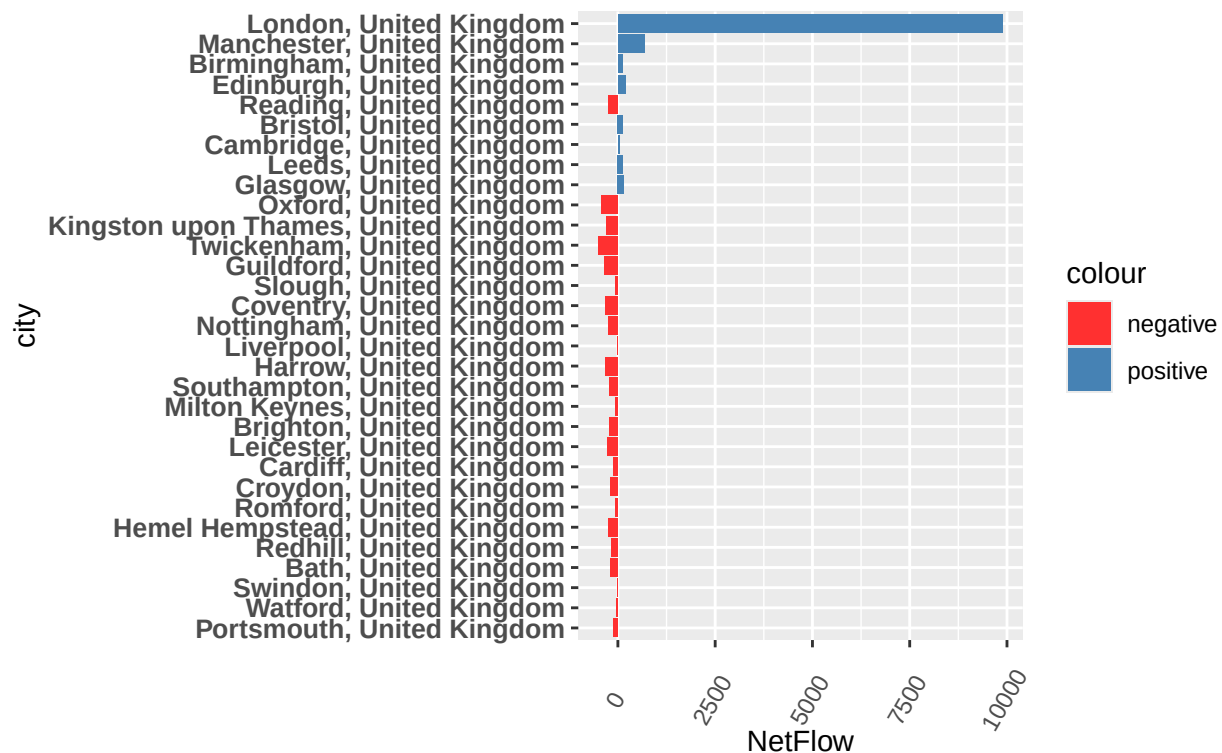
```
Top_outUK + coord_flip()
```



```
# 3.a) NET flow in Top DESTINATION
Top_netUK <- aggregateByCityUK %>%
  filter(country == "United Kingdom") %>% # filter by country = UK
  top_n(30, NumInflow)
Top_netUK$city = with(Top_netUK, reorder(city, NumInflow)) # reorder Levels by Var

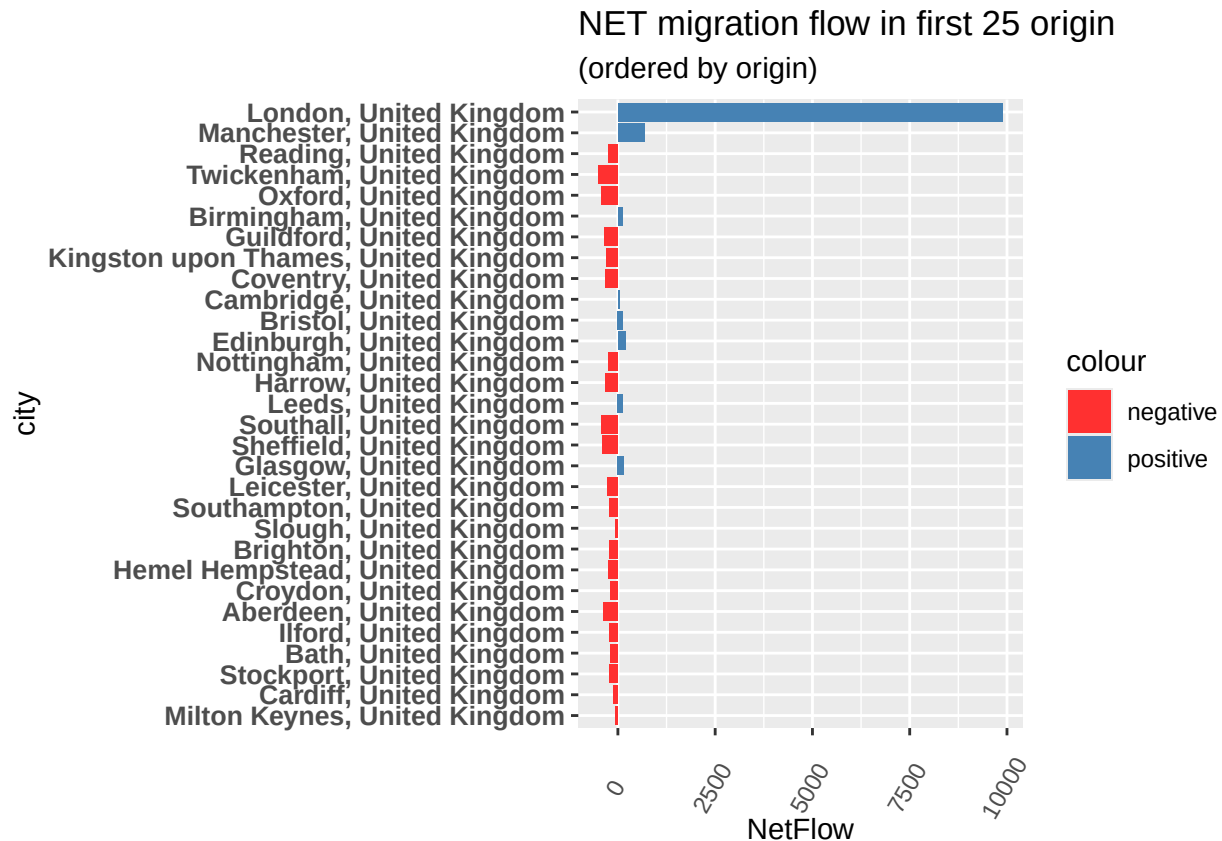
Top_netUK <- ggplot(data = Top_netUK, aes(city, NetFlow)) +
  geom_bar(stat = "identity", position="identity", aes(fill = colour)) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10, face="bold"),
  scale_fill_manual(values=c(positive="steelblue", negative="firebrick1")) +
  labs(title="NET migration flow in first 25 destination", subtitle="(ordered by destination)")
Top_netUK + coord_flip()
```

NET migration flow in first 25 destination
(ordered by destination)



```
# 3.b) NET flow in Top ORIGIN
Top_netUK <- aggreByCityUK %>%
  filter (country == "United Kingdom") %>% # filter by country = UK
  top_n(30, NumOutflow)
Top_netUK$city = with(Top_netUK, reorder(city, NumOutflow)) # reorder Levels by Var

Top_netUK <- ggplot(data = Top_netUK, aes(city, NetFlow)) +
  geom_bar(stat = "identity", position="identity", aes(fill = colour)) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10, face="bold")) +
  scale_fill_manual(values=c(positive="steelblue", negative="firebrick1")) +
  labs(title="NET migration flow in first 25 origin", subtitle="(ordered by origin)")
Top_netUK + coord_flip()
```



- **INSIGHTs:**
 - Contrary to US, London is a definitive outlier

Major 20 (AUSTRALIA) cities TO / FROM / NET migration

```
# 1) In Flow in Top DESTINATION

aggreByCityAustr <- aggreByCity2 %>%
  filter (country == "Australia")

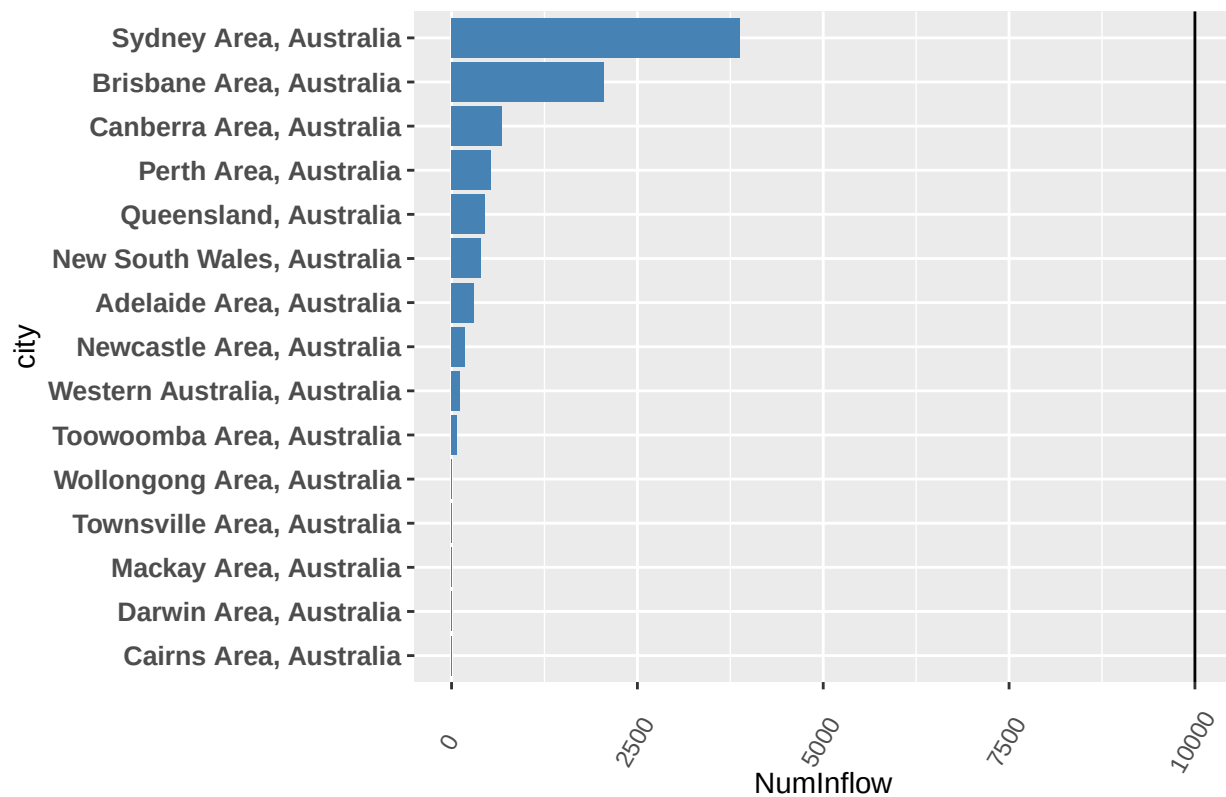
Top_inAustr <- aggreByCityAustr %>%
  top_n(30, NumInflow)

Top_inAustr$city = with(Top_inAustr, reorder(city, NumInflow)) # reorder Levels by Var

Top_inAustr <- ggplot(data = Top_inAustr, aes(city, NumInflow)) +
  geom_bar(stat = "identity", position="identity", fill = "steelblue") +
  geom_hline(yintercept=10000, color = "black", size=0.5) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10, face="bold"),
  labs(title="IN migration flow in first 15 destination")

Top_inAustr + coord_flip()
```

IN migration flow in first 15 destination

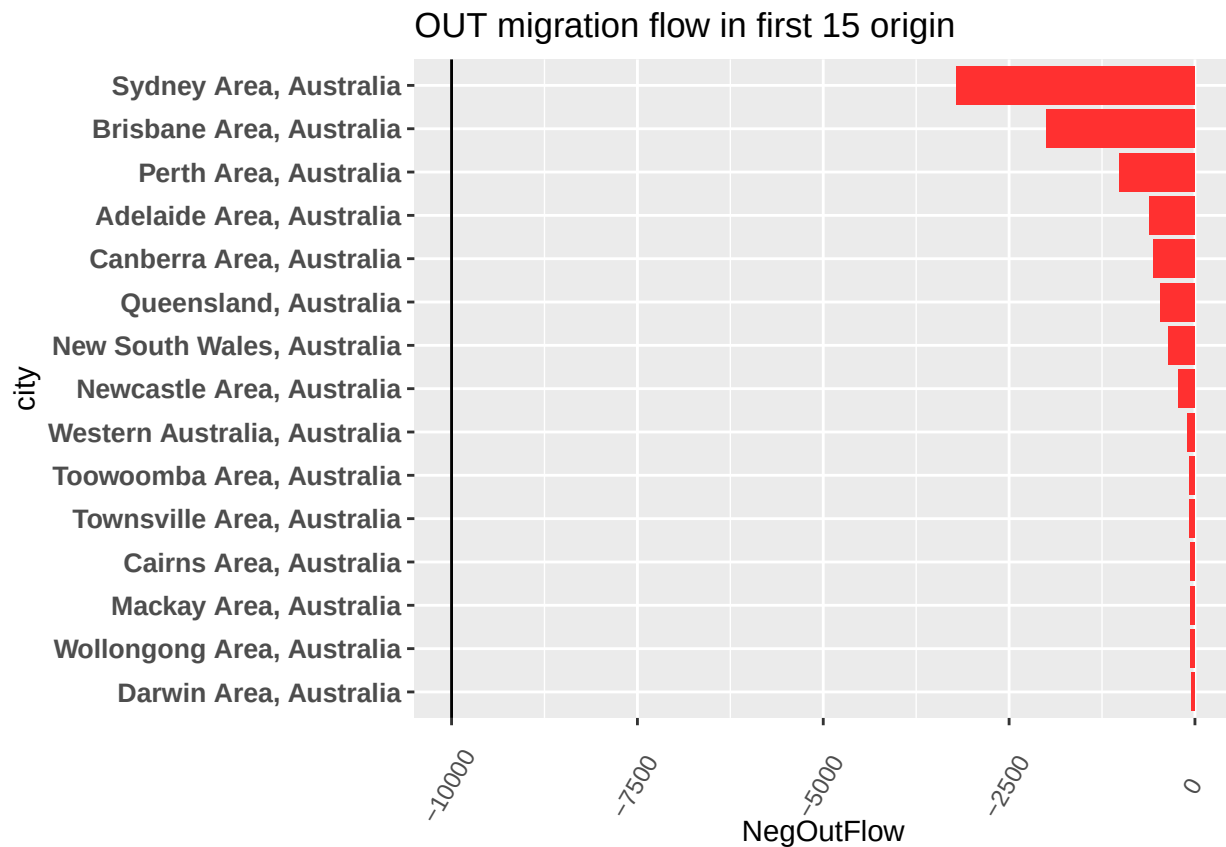


2) Out Flow in Top ORIGIN

```
Top_outAustr <- aggreByCityAustr %>%
  filter (country == "Australia") %>% # filter by country = Austr
  top_n(30, NumOutflow)
Top_outAustr$city = with(Top_outAustr, reorder(city, NumOutflow)) # reorder Levels by Var
```

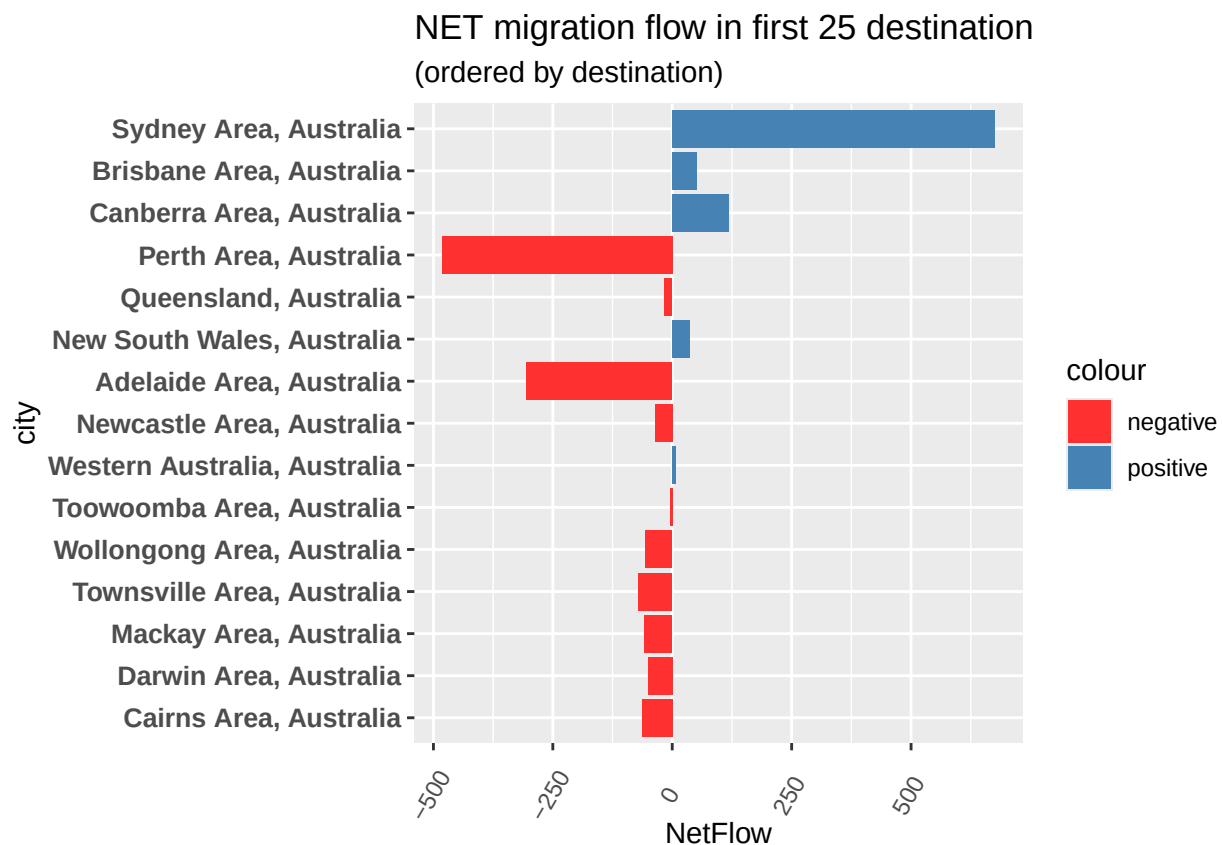
```
Top_outAustr <- ggplot(data = Top_outAustr, aes(city, NegOutFlow)) +
  geom_bar(stat = "identity", position="identity", fill = "firebrick1") +
  geom_hline(yintercept=-10000, color = "black", size=0.5) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10, face="bold"))
labs(title="OUT migration flow in first 15 origin")
```

```
Top_outAustr + coord_flip()
```



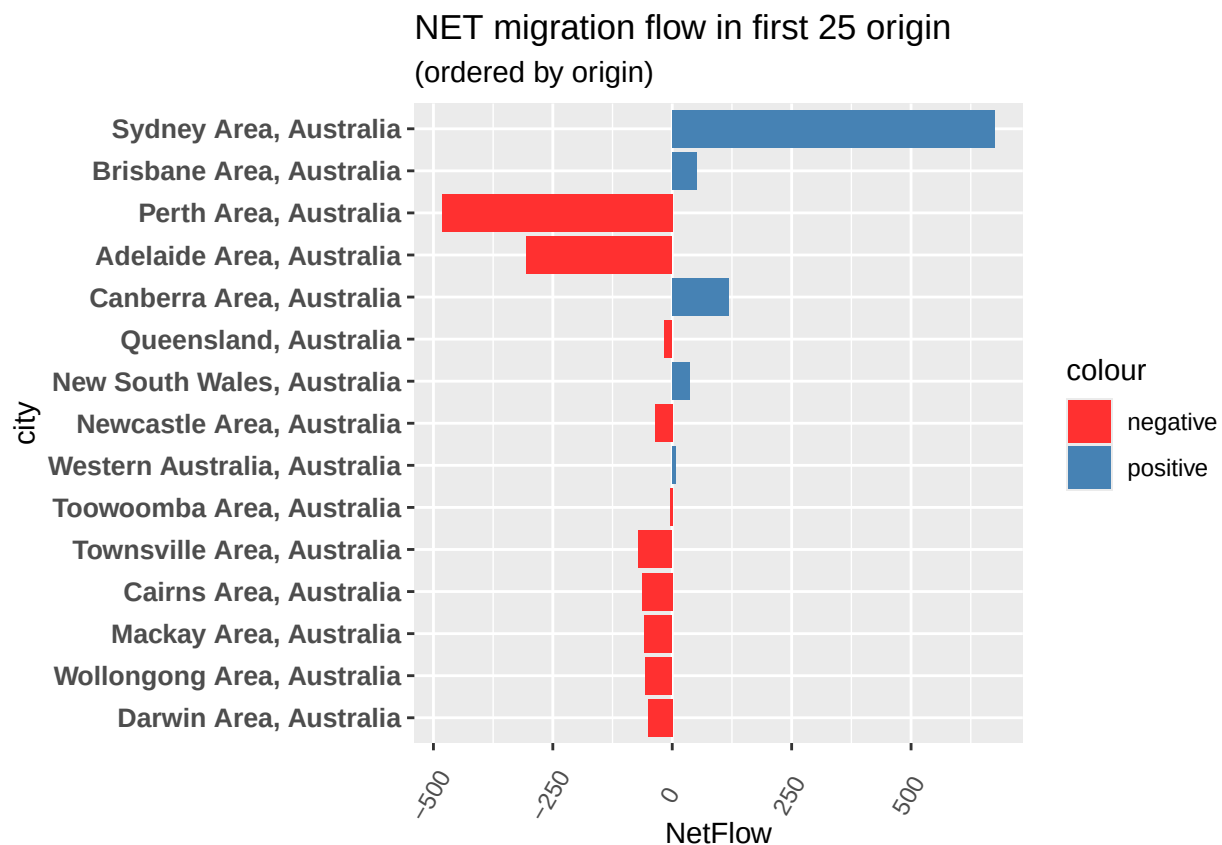
```
# 3.a) NET flow in Top DESTINATION
Top_netAustr <- aggreByCityAustr %>%
  filter (country == "Australia") %>% # filter by country = Austr
  top_n(30, NumInflow)
Top_netAustr$city = with(Top_netAustr, reorder(city,NumInflow)) # reorder Levels by Var

Top_netAustr <- ggplot(data = Top_netAustr,aes(city, NetFlow)) +
  geom_bar(stat = "identity", position="identity",aes(fill = colour)) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10,face="bold"),
  scale_fill_manual(values=c(positive="steelblue",negative="firebrick1")) +
  labs(title="NET migration flow in first 25 destination", subtitle="(ordered by destination)")
Top_netAustr + coord_flip()
```



```
# 3.b) NET flow in Top ORIGIN
Top_netAustr <- aggreByCityAustr %>%
  filter (country == "Australia") %>% # filter by country = Austr
  top_n(30, NumOutflow)
Top_netAustr$city = with(Top_netAustr, reorder(city, NumOutflow)) # reorder Levels by Var

Top_netAustr <- ggplot(data = Top_netAustr, aes(city, NetFlow)) +
  geom_bar(stat = "identity", position="identity", aes(fill = colour)) +
  theme(axis.text.x = element_text(angle=60, vjust=0.3), axis.text.y = element_text(size=10, face="bold")) +
  scale_fill_manual(values=c(positive="steelblue", negative="firebrick1")) +
  labs(title="NET migration flow in first 25 origin", subtitle="(ordered by origin)")
Top_netAustr + coord_flip()
```



4 BIVARIATE MEASURES OF ASSOCIATION

4.1 INflow by city vs highest degree

Is there any relation between where they choose to go and the highest degree they have? I use plots to check distributions type of DEGREE (Y) conditional on the city of destination (X)

- I check for :
 - *Existence*
 - *strnght*
 - *patterns* / direction

Greater New York City Area San Francisco Bay Area Washington D.C. Metro Area

Manchester, United Kingdom London, United Kingdom Birmingham, United Kingdom

Sydney Area, Australia Brisbane Area, Australia Canberra Area, Australia

```
# 1) attempt for simplicity I select top 3 per country
```

```
#top3 <- both %>% filter(DESTINATION_REGION == "Greater New York City Area" | DESTINATION_REGION == "San Francisco Bay Area" | DESTINATION_REGION == "Washington D.C. Metro Area")
```

```
# 2) attempt
```

```
# top_subs <- subset(both, (DESTINATION_REGION == 'Greater New York City Area' | DESTINATION_REGION == 'San Francisco Bay Area' | DESTINATION_REGION == 'Washington D.C. Metro Area'))
```

```
# 3) attempt
```

```
# When subsetting with [ names are always matched exactly
```

```

# z <- c(abc = 1, def = 2)
# z[c("a", "d")]

top_dest <- c("Greater New York City Area" , "San Francisco Bay Area", "Washington D.C. Metro Area" ,

# both_top <- both[as.character( both$DESTINATION_REGION %in% top_dest), drop = T]

# data[data$Code %in% selected,]
# both_top <- both[both$DESTINATION_REGION %in% top_dest]
# both_top <- both[as.character( both$DESTINATION_REGION %in% top_dest), drop = TRUE]

# 4) attempts
# data[data$Code == "A" | data$Code == "B", ]
top3cit <- both[both$DESTINATION_REGION == "Greater New York City Area" | both$DESTINATION_REGION == "S

# 5) attempts
# top_dest <- c("Greater New York City Area" , "San Francisco Bay Area", "Washington D.C. Metro Area"

# top3cit_2 <- both[both$DESTINATION_REGION %in% top_dest, , drop =TRUE ]

# mutate (Sign = ifelse(NetFlow > 0, "Positive", "Negative"))

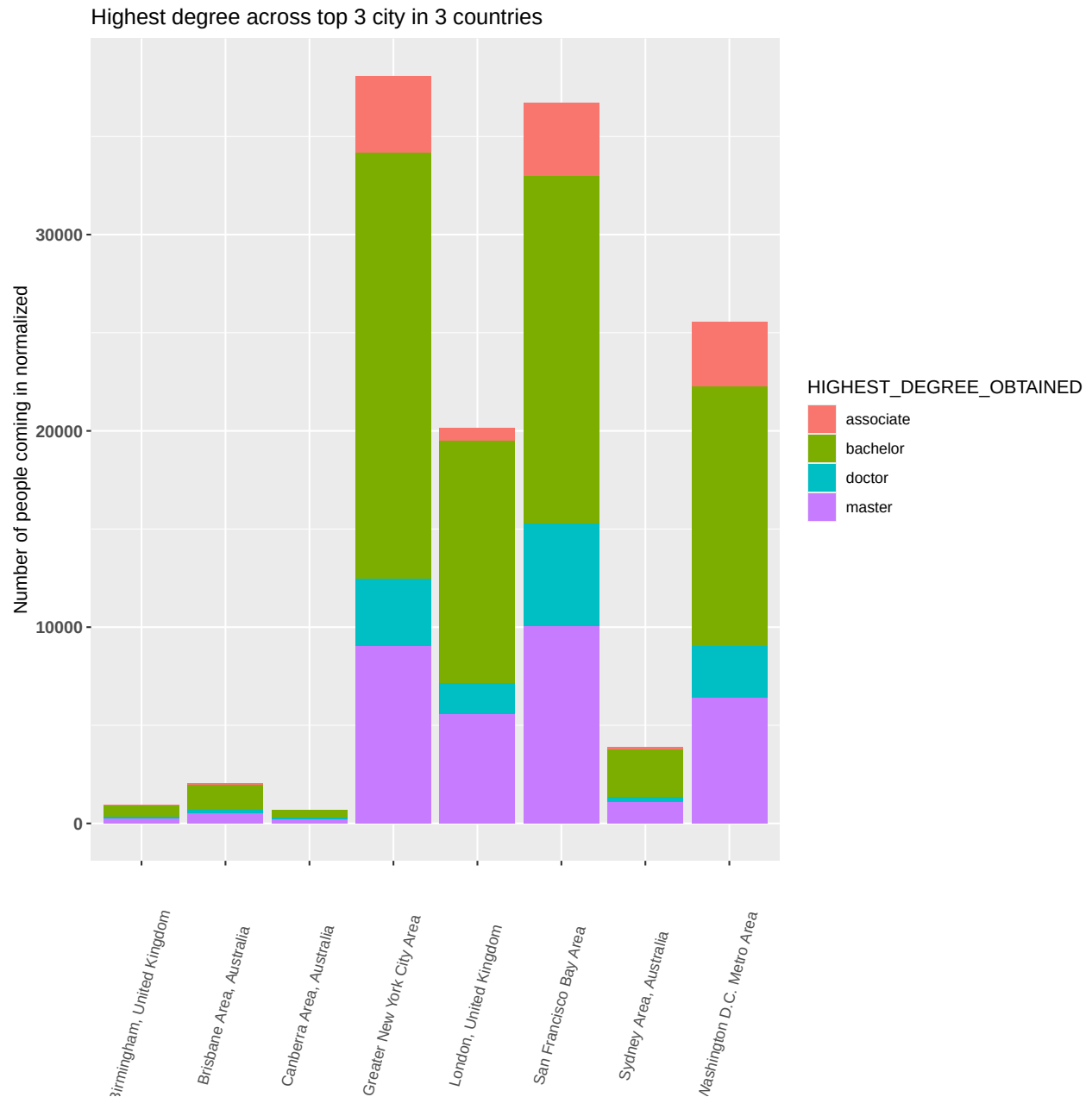
# Explore

# mosaicplot(table(top3cit$DESTINATION_REGION, top3cit$HIGHEST_DEGREE_OBTAINED), ylab = "Political Part

# qplot
qplot(x = DESTINATION_REGION, data = top3cit, fill = HIGHEST_DEGREE_OBTAINED, geom = "bar") +
theme(axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10,face="bold") )
labs(title="Highest degree across top 3 city in 3 countries" , x="", y="Number of people coming in no

## Warning: `qplot()` was deprecated in ggplot2 3.4.0.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

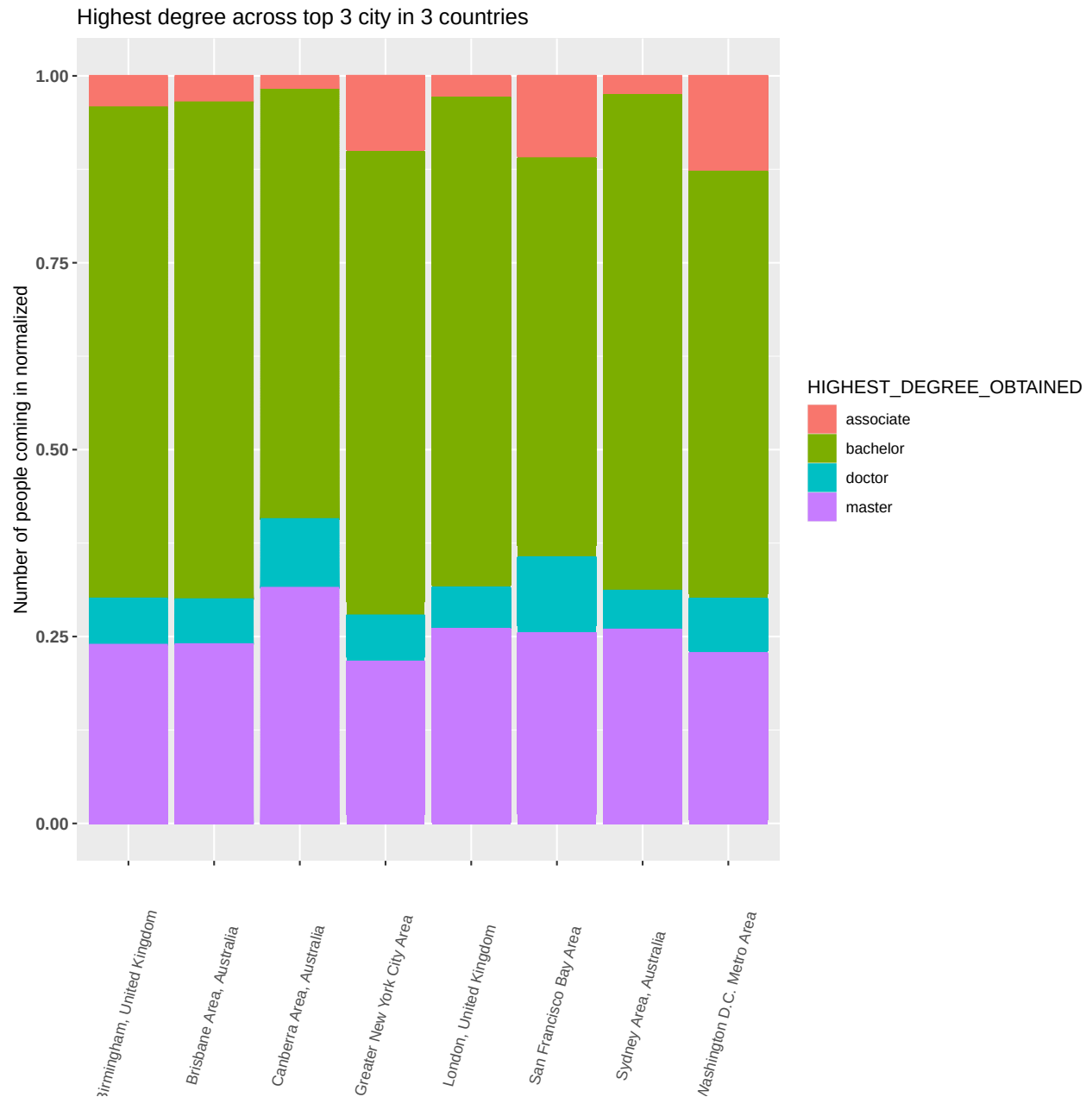
```



```
# ggplot

#ggplot(data = Best_out,aes(cityD, NegOutFlow)) +
# geom_bar(stat = "identity", position="identity", fill = "firebrick1") +
# geom_hline(yintercept=-10000, color = "black", size=0.5) +
# theme(axis.text.x = element_text(angle=60, vjust=0.3)) +
# labs(title="OUT migration flow in first 15 destination")

ggplot(top3cit, aes(x=DESTINATION_REGION, y=NEW_MEM_ID, fill=HIGHEST_DEGREE_OBTAINED)) +
  geom_bar(aes(colour =HIGHEST_DEGREE_OBTAINED),stat="identity", position = "fill") +
  theme(axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10,face="bold")) +
  labs(title="Highest degree across top 3 city in 3 countries" , x="", y="Number of people coming in no")
```



- **INSIGHTs:**

- Intuitively, it would seem San Francisco (followed by Canberra) attracts the highest amount of doctors (Canberra also the highest group with master)
- NY, San Francisco and Washington DC seem to receive many with “Associate” level: either young people go there to look for their first job

```
# cramer v degree X CITY OF DESTINATION (top 3)
x<- top3cit$DESTINATION_REGION
y<- top3cit$HIGHEST_DEGREE_OBTAINED

cv.test = function(x,y) {
  CV = sqrt(chisq.test(x, y, correct=FALSE)$statistic /
    (length(x) * (min(length(unique(x)),length(unique(y))) - 1)))
  print.noquote("Cramér V / Phi:")
}
```

```

    return(as.numeric(CV))
}

with(top3cit, cv.test(x, y)) # [1] Cramér V / Phi: 0.09052046

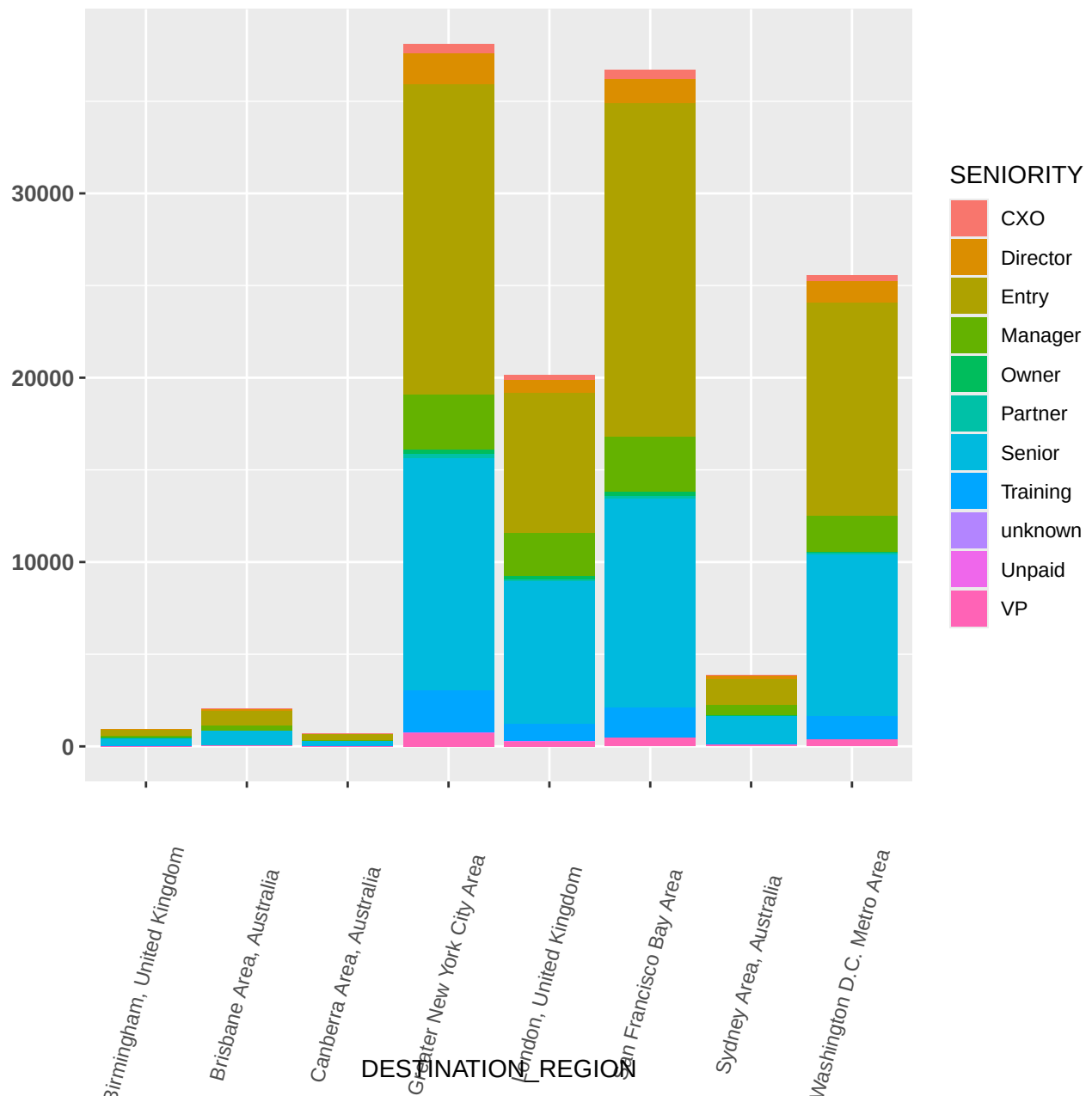
## [1] Cramér V / Phi:
## [1] 0.09052046

#
# mosaicplot(table(top3cit$DESTINATION_REGION, top3cit$SENIORITY), ylab = "Political Party", xlab = "Ta

# qplot
qplot(x = DESTINATION_REGION, data = top3cit, fill = SENIORITY, geom = "bar") +
theme(axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10,face="bold"))
labs(title="SENIORITY across top 3 city in 3 countries")

```

SENIORITY across top 3 city in 3 countries



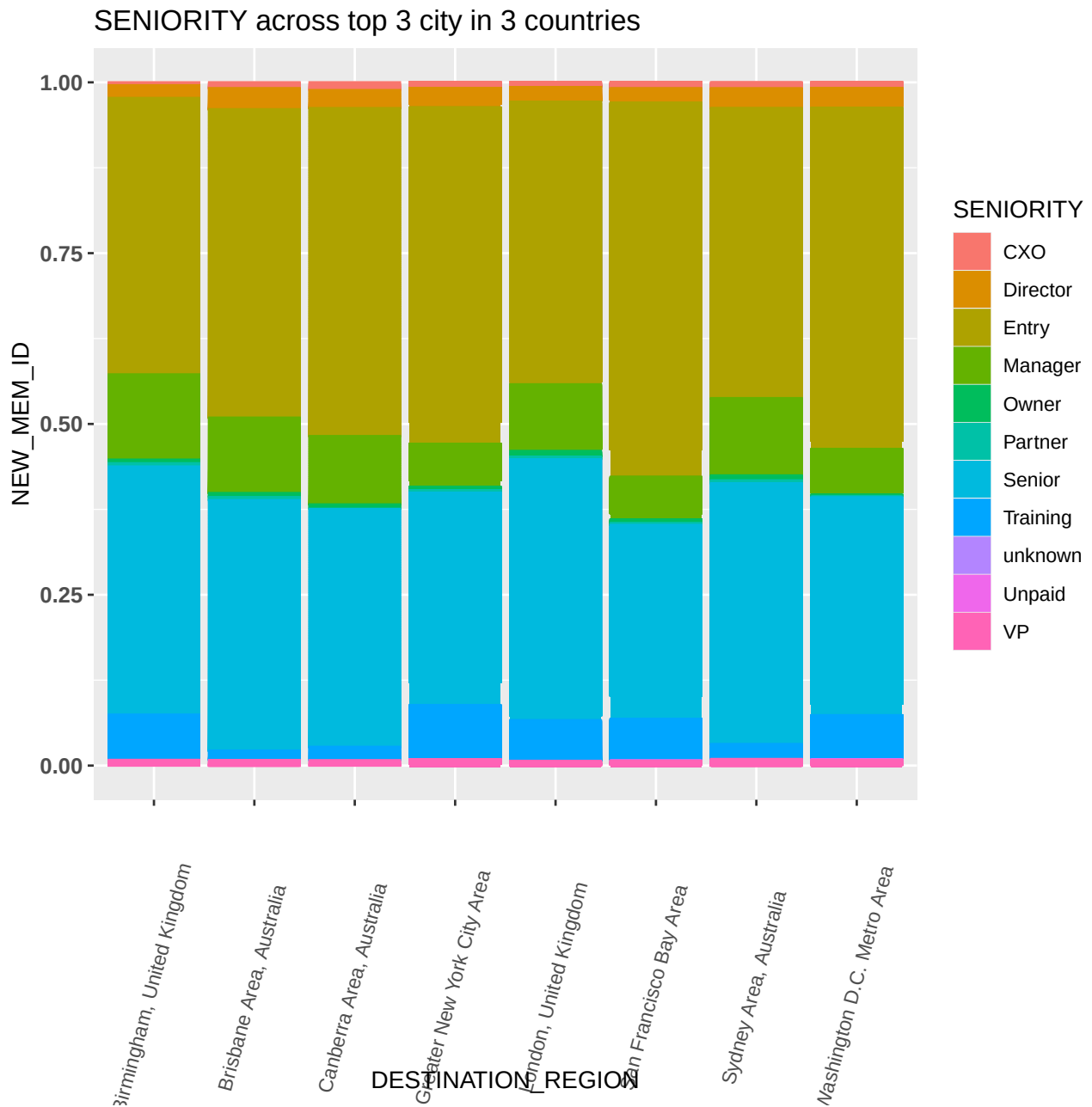
```
# Test statistic (assuming independence)
chisq <- chisq.test(x = table(top3cit$DESTINATION_REGION, top3cit$SENIORITY), correct = FALSE)

## Warning in chisq.test(x = table(top3cit$DESTINATION_REGION, top3cit$SENIORITY),
## : Chi-squared approximation may be incorrect

chisq

##
## Pearson's Chi-squared test
##
## data:  table(top3cit$DESTINATION_REGION, top3cit$SENIORITY)
## X-squared = NaN, df = 2810, p-value = NA
```

```
# ggplot
ggplot(top3cit, aes(x=DESTINATION_REGION, y=NEW_MEM_ID, fill=SENIORITY)) + geom_bar(aes(colour =SENIORITY))
theme(axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10,face="bold"))
labs(title="SENIORITY across top 3 city in 3 countries")
```



* **INSIGHTS:** + Intuitively, it would seem San Francisco (followed by Canberra) attracts the highest amount of doctors (Canberra also the highest group with master) + NY, San Francisco and Washington DC seem to receive many with “Associate” level: either young people go there to look for their first job

```
# cramer v SENIORITY X CITY OF DESTINATION (top 3)
x<- top3cit$DESTINATION_REGION
y<- top3cit$SENIORITY

cv.test = function(x,y) {
```

```

CV = sqrt(chisq.test(x, y, correct=FALSE)$statistic /
  (length(x) * (min(length(unique(x)),length(unique(y))) - 1)))
print.noquote("Cramér V / Phi:")
return(as.numeric(CV))
}

with(top3cit, cv.test(x, y)) # [1] Cramér V / Phi: 0.04360073

## Warning in chisq.test(x, y, correct = FALSE): Chi-squared approximation may be
## incorrect

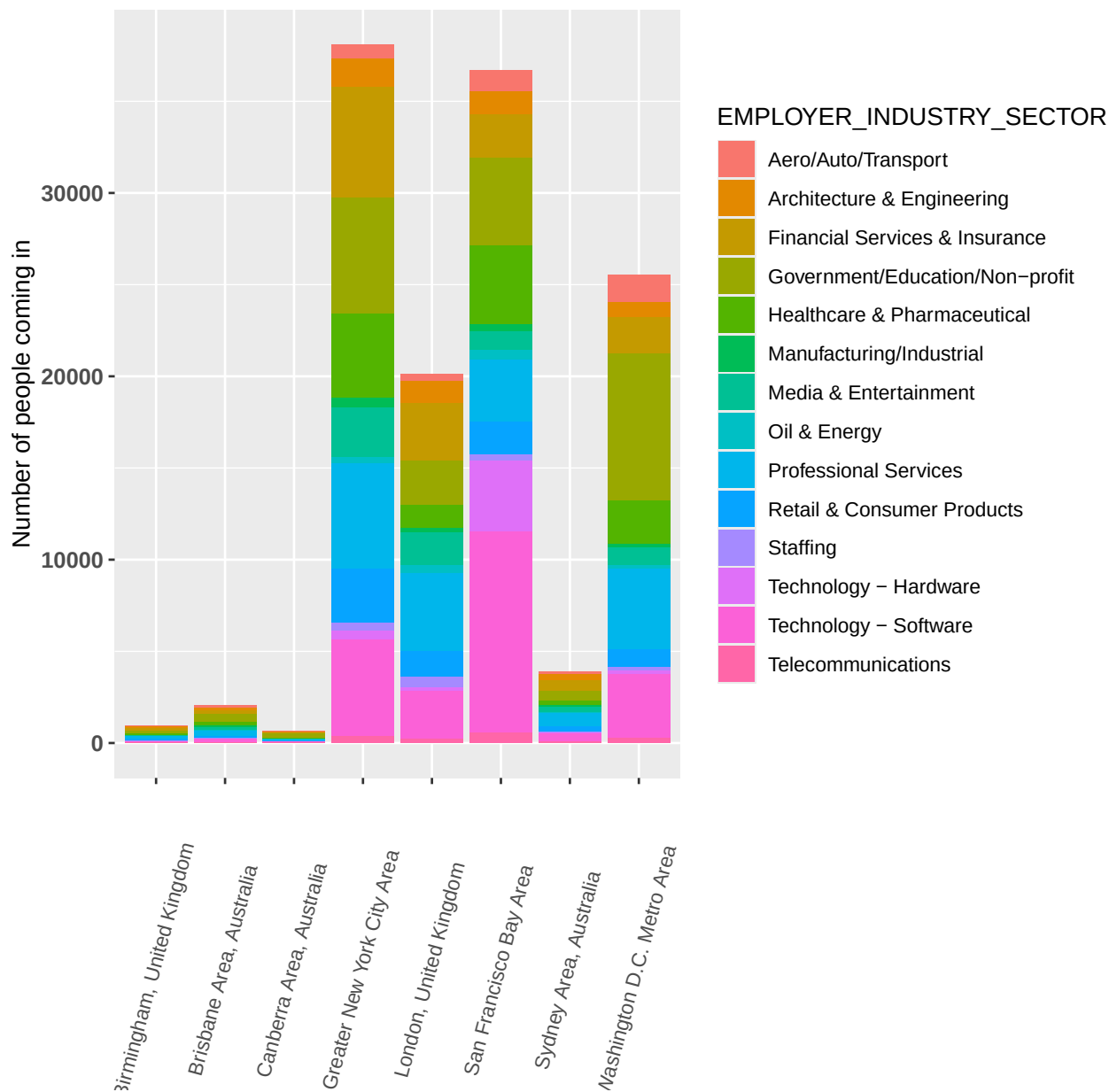
## [1] Cramér V / Phi:
## [1] 0.04360073

#
# mosaicplot(table(top3cit$DESTINATION_REGION, top3cit$EMPLOYER_INDUSTRY_SECTOR), ylab = "Political Par

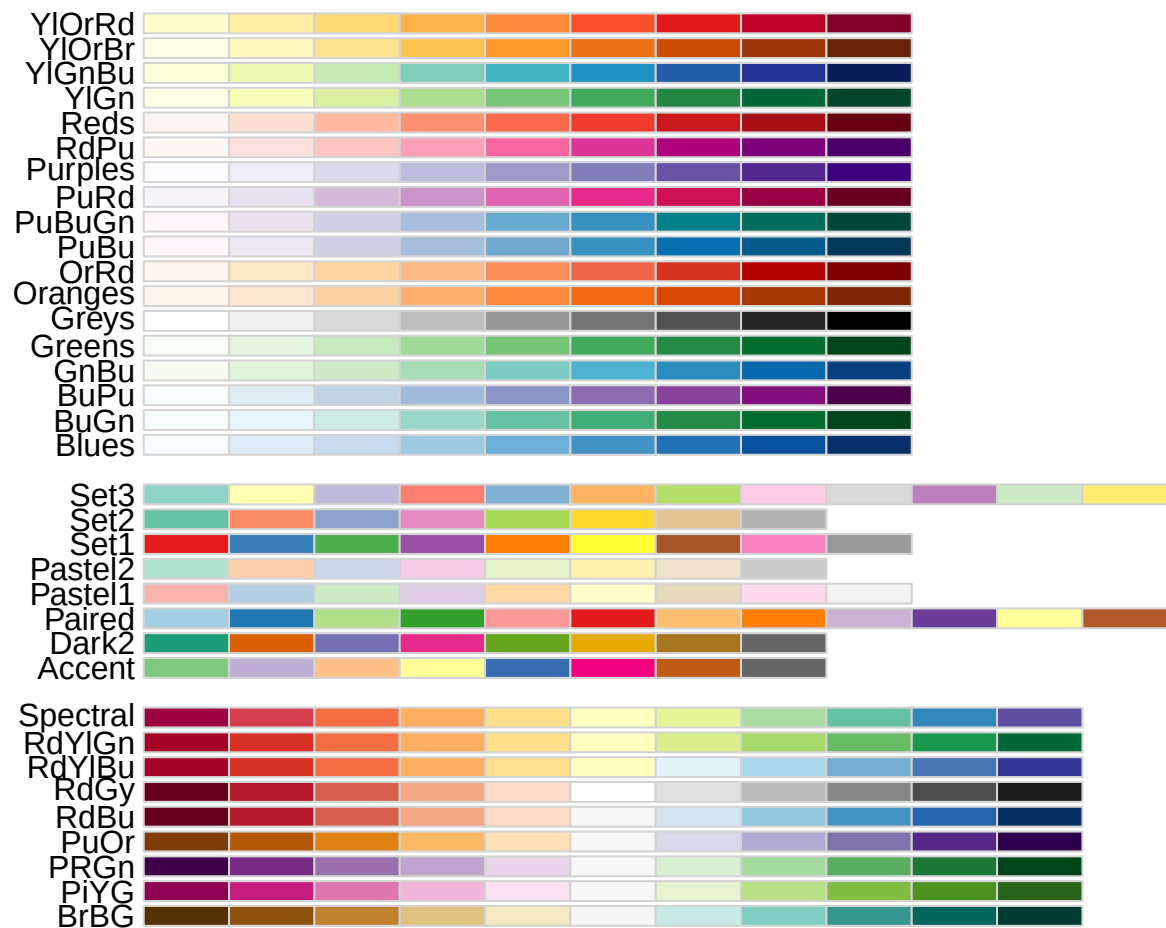
# qplot
qplot(x = DESTINATION_REGION, data = top3cit, fill = EMPLOYER_INDUSTRY_SECTOR, geom = "bar") +
theme(axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10,face="bold"))
labs(title="EMPLOYER_INDUSTRY_SECTOR across top 3 city in 3 countries", x="", y="Number of people comin

```

EMPLOYER_INDUSTRY_SECTOR across top 3 city in 3 countries

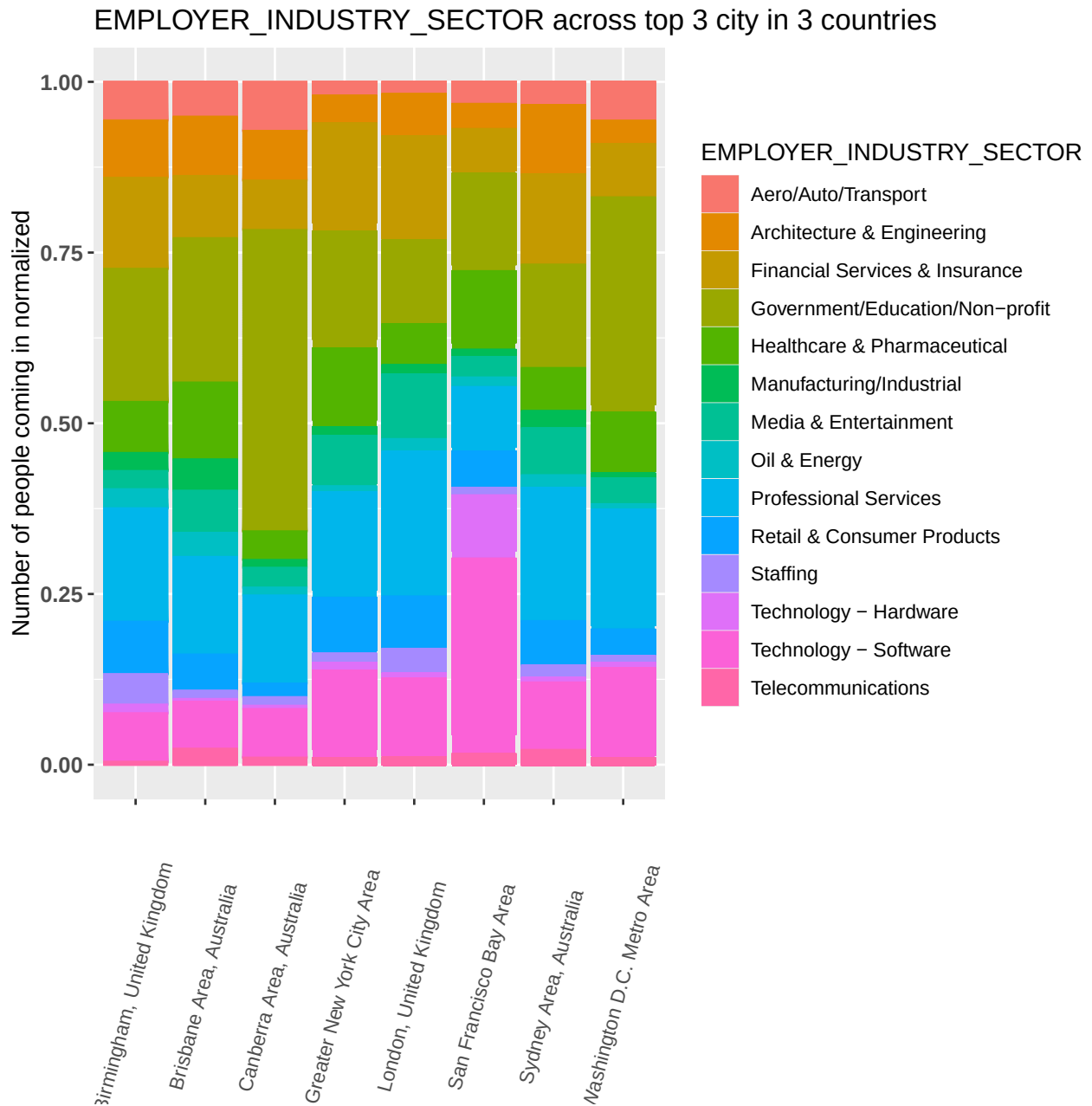


```
# ggplot
library(RColorBrewer)
display.brewer.all()
```



```
ggplot(top3cit, aes(x=DESTINATION_REGION, y=NEW_MEM_ID, fill=EMPLOYER_INDUSTRY_SECTOR)) +
  geom_bar(aes(colour =EMPLOYER_INDUSTRY_SECTOR),stat="identity", position = "fill") +
  theme(legend.position = "right",axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(angle=0, vjust=0.5))

# scale_fill_brewer(palette = "Dark2") +
# scale_fill_grey() +
labs(title="EMPLOYER_INDUSTRY_SECTOR across top 3 city in 3 countries", x="", y="Number of people com")
```



• **INSIGHTS:**

- Intuitively, share of immigrants by sector seems to vary a lot in different cities
- Extremely high number in Govt/ Educ/ Non Profit in Canberra & Washington
- Extremely high number in Software + Hardware Technology in San Francisco
- Extremely high number in Govt/ Educ/ Non Profit + Financial services also in NY it would seem San Francisco (followed by Canberra) attracts the highest amount of doctors (Canberra also the highest group with master)
- NY, San Francisco and Washington DC seem to receive many with “Associate” level: either young people go there to look for their first job

CHECK A COUPLE OF CITIES FOR SECTOR X SENIORITY

```
freq_OrigDegree <- both %>%
  group_by(both[,7],both[,2]) %>%
  summarise (n = n()) %>%
  mutate(freq = n / sum(n)) %>%
  mutate(rel.freq = paste0(round(100 * n/sum(n), 2), "%"))

## `summarise()` has grouped output by 'both[, 7]'. You can override using the
## `.groups` argument.
freq_OrigDegree # US has a significantly higher # of Associates leaving (18% vs 3% and 4%)

## # A tibble: 12 x 5
## # Groups:   both[, 7] [3]
##   `both[, 7]`   `both[, 2]`      n    freq rel.freq
##   <fct>        <fct>    <int> <dbl> <chr>
## 1 Australia    associate     255 0.0286 2.86%
## 2 Australia    bachelor    5632 0.633 63.27%
## 3 Australia    doctor       611 0.0686 6.86%
## 4 Australia    master      2404 0.270 27.01%
## 5 United Kingdom associate    1312 0.0358 3.58%
## 6 United Kingdom bachelor    22107 0.604 60.38%
## 7 United Kingdom doctor      3533 0.0965 9.65%
## 8 United Kingdom master      9663 0.264 26.39%
## 9 United States associate    79505 0.185 18.5%
## 10 United States bachelor   225213 0.524 52.4%
## 11 United States doctor    36358 0.0846 8.46%
## 12 United States master    88723 0.206 20.64%
```

Cramer's V a measure of association for nominal variables. Effectively it is the Pearson chi-square statistic rescaled to have values between 0 and 1, as follows:

$$\phi_c = \sqrt{\frac{\chi^2}{N * (\min(ncols, nrows) - 1)}}$$

where χ^2 is the Pearson chi-square, nobs represents the number of observations included in the table, and where ncols and nrows are the number of rows and columns in the table, respectively.

For a 2 by 2 table, of course, this is just the square root of chi-square divided by the number of observations, which is also known as the ϕ coefficient.

Cramer's V varies from 0 (corresponding to no association between the variables) to 1 (complete association) and can reach 1 only when the two variables are equal to each other

```
# cramer v   Sector X CITY OF DESTINATION (top 3)
x<- top3cit$DESTINATION_REGION
y<- top3cit$EMPLOYER_INDUSTRY_SECTOR

cv.test = function(x,y) {
  CV = sqrt(chisq.test(x, y, correct=FALSE)$statistic /
    (length(x) * (min(length(unique(x)),length(unique(y))) - 1)))
  print.noquote("Cramér V / Phi:")
  return(as.numeric(CV))
}

with(top3cit, cv.test(x, y)) # [1] Cramér V / Phi: 0.1605205
```

```
## [1] Cramér V / Phi:
```

```
## [1] 0.1605205
```

```
# how about across all cities? (lower)
```

```
x<- both$DESTINATION_REGION
```

```
y<- both$EMPLOYER_INDUSTRY_SECTOR
```

```
cv.test = function(x,y) {
```

```
  CV = sqrt(chisq.test(x, y, correct=FALSE)$statistic /  
    (length(x) * (min(length(unique(x)),length(unique(y))) - 1)))
```

```
  print.noquote("Cramér V / Phi:")
```

```
  return(as.numeric(CV))
```

```
}
```

```
with(both, cv.test(x, y)) # [1] Cramér V / Phi: 0.1264835
```

```
## Warning in chisq.test(x, y, correct = FALSE): Chi-squared approximation may be
```

```
## incorrect
```

```
## [1] Cramér V / Phi:
```

```
## [1] 0.1264835
```

position

```
#
```

```
# mosaicplot(table(top3cit$DESTINATION_REGION, top3cit$POSITION_FUNCTION), ylab = "Political Party", xlab = "Position Function")
```

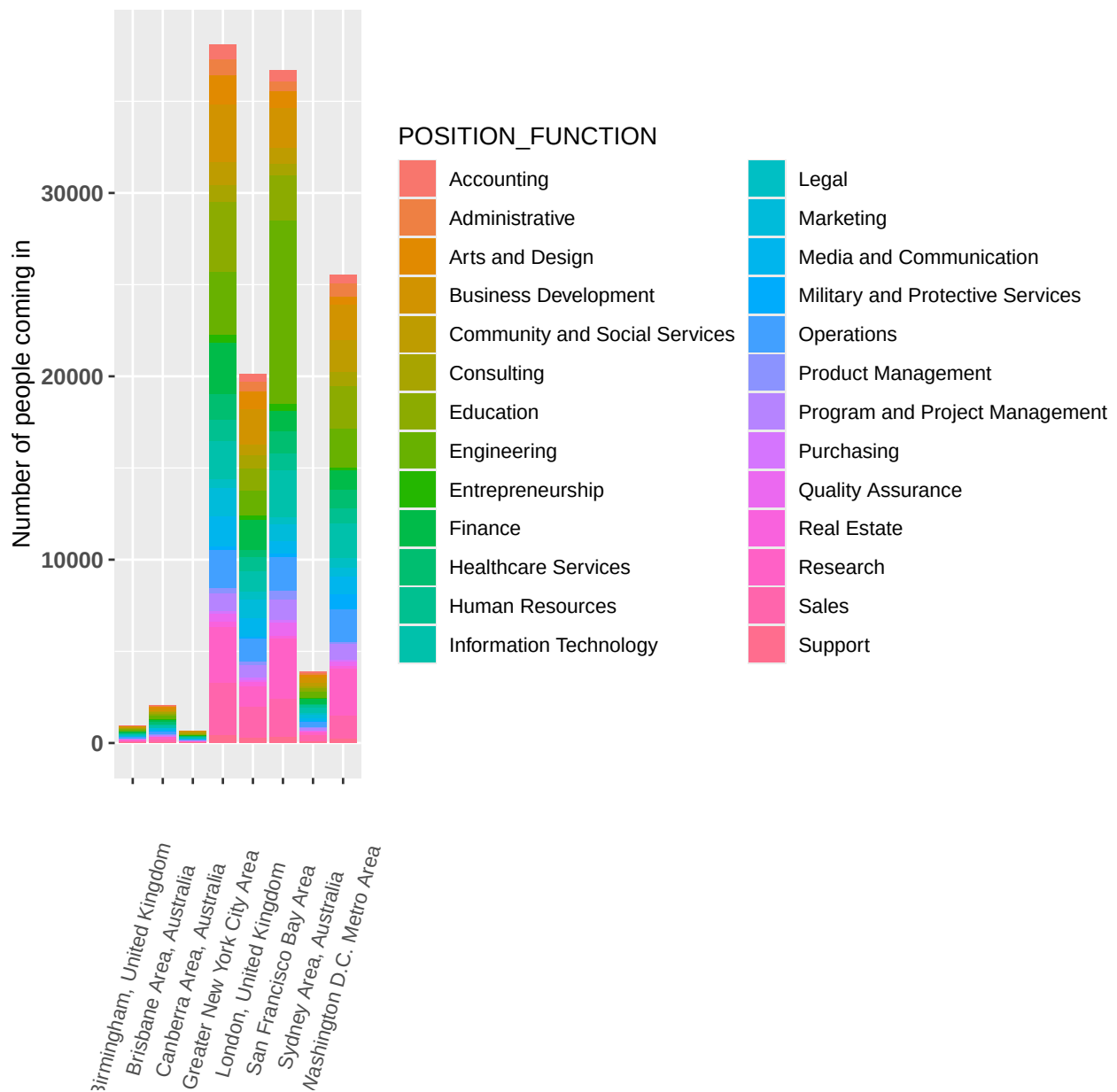
```
# qplot
```

```
qplot(x = DESTINATION_REGION, data = top3cit, fill = POSITION_FUNCTION, geom = "bar") +
```

```
theme(axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10,face="bold"))
```

```
  labs(title="POSITION_FUNCTION across top 3 city in 3 countries" , x="", y="Number of people coming in")
```

POSITION_FUNCTION across top 3 city in 3 countries



```
# Test statistic (assuming independence)
chisq <- chisq.test(x = table(top3cit$DESTINATION_REGION, top3cit$POSITION_FUNCTION), correct = FALSE)

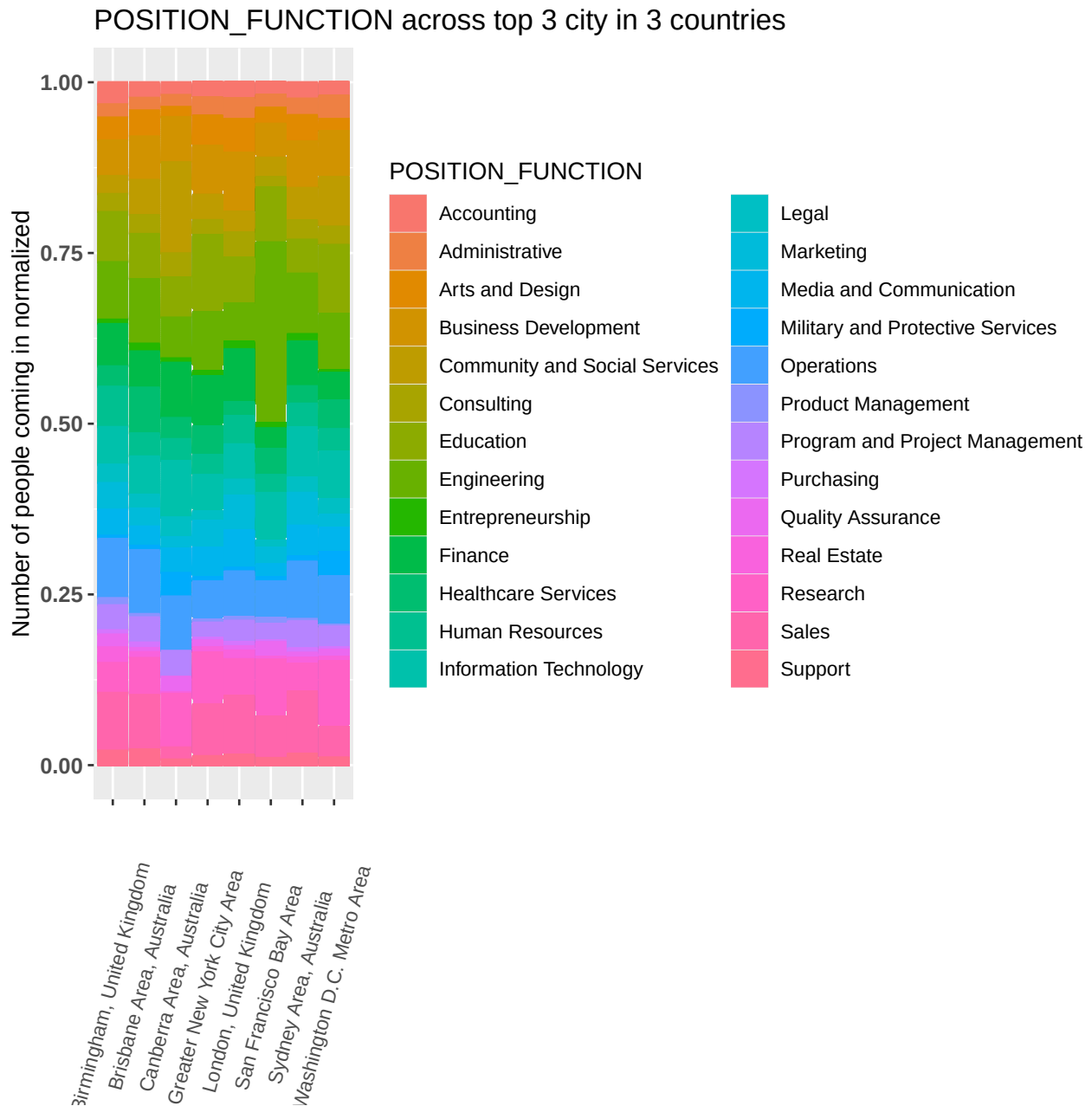
## Warning in chisq.test(x = table(top3cit$DESTINATION_REGION,
## top3cit$POSITION_FUNCTION), : Chi-squared approximation may be incorrect

chisq

##
## Pearson's Chi-squared test
##
## data:  table(top3cit$DESTINATION_REGION, top3cit$POSITION_FUNCTION)
## X-squared = NaN, df = 7025, p-value = NA
```

```
# ggplot
```

```
ggplot(top3cit, aes(x=DESTINATION_REGION, y=NEW_MEM_ID, fill=POSITION_FUNCTION), colour="black") +  
  geom_bar(aes(colour =POSITION_FUNCTION),stat="identity", position = "fill" ) +  
  theme(axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10,face="bold"))  
  labs(title="POSITION_FUNCTION across top 3 city in 3 countries", x="", y="Number of people coming in")
```



```
# cramer v POSITION_FUNCTION X CITY OF DESTINATION (top 3)
```

```
x<- top3cit$DESTINATION_REGION
```

```
y<- top3cit$POSITION_FUNCTION
```

```
cv.test = function(x,y) {
```

```
  CV = sqrt(chisq.test(x, y, correct=FALSE)$statistic /  
    (length(x) * (min(length(unique(x)),length(unique(y))) - 1)))
```

```

print.noquote("Cramér V / Phi:")
return(as.numeric(CV))
}

with(top3cit, cv.test(x, y)) # [1] Cramér V / Phi: [1] 0.1318759

## Warning in chisq.test(x, y, correct = FALSE): Chi-squared approximation may be
## incorrect
## [1] Cramér V / Phi:
## [1] 0.1318759

# how about across all cities? (lower)
x<- both$DESTINATION_REGION
y<- both$POSITION_FUNCTION

cv.test = function(x,y) {
  CV = sqrt(chisq.test(x, y, correct=FALSE)$statistic /
    (length(x) * (min(length(unique(x)),length(unique(y))) - 1)))
  print.noquote("Cramér V / Phi:")
  return(as.numeric(CV))
}

with(both, cv.test(x, y)) # [1] Cramér V / Phi: [1] 0.07209771

## Warning in chisq.test(x, y, correct = FALSE): Chi-squared approximation may be
## incorrect
## [1] Cramér V / Phi:
## [1] 0.07209771

```

5. DOMESTIC MIGRATION

comparative analysis of domestic / patterns of internal migration in each country

I will focus on the sector per city which seems the most significant bivariate association

```

# select only USA
both_USA <- both %>% filter(SOURCE_COUNTRY=="United States" & DESTINATION_COUNTRY=="United States")

# select only Australia
both_AUS <- both %>% filter(SOURCE_COUNTRY=="Australia" & DESTINATION_COUNTRY=="Australia")

# select only UK
both_UK <- both %>% filter(SOURCE_COUNTRY=="United Kingdom" & DESTINATION_COUNTRY=="United Kingdom")

```

construct different samples

```

both_USA_N <- both_USA %>% group_by(DESTINATION_REGION) %>% summarise(numIMM = n())
both_USA_N

```

construct USA internal sub-sample (for simplicity)

```
## # A tibble: 212 x 2
##   DESTINATION_REGION      numIMM
##   <fct>                  <int>
## 1 Abilene, Texas Area      59
## 2 Albany, New York Area   1165
## 3 Albuquerque, New Mexico Area 335
## 4 Allentown, Pennsylvania Area 728
## 5 Anchorage, Alaska Area   79
## 6 Asheville, North Carolina Area 205
## 7 Athens, Georgia Area    311
## 8 Auburn, Alabama Area     60
## 9 Augusta, Georgia Area   201
## 10 Austin, Texas Area     8401
## # i 202 more rows

some_USA <- both_USA[both_USA$DESTINATION_REGION == "Greater New York City Area" | both_USA$DESTINATION_REGION == "Green Bay, Wisconsin Area" | both_USA$DESTINATION_REGION == "San Francisco Bay Area" | both_USA$DESTINATION_REGION == "Washington D.C. Metro Area" | both_USA$DESTINATION_REGION == "Greater Los Angeles Area" | both_USA$DESTINATION_REGION == "Greater Boston Area" | both_USA$DESTINATION_REGION == "Greater Chicago Area" | both_USA$DESTINATION_REGION == "Greater Seattle Area" | both_USA$DESTINATION_REGION == "Other", ]

summary(some_USA)
```

NEW_MEM_ID	HIGHEST_DEGREE_OBTAINED	SENIORITY
Min. : 4	associate:25165	Entry :84953
1st Qu.:112192	bachelor :94375	Senior :55988
Median :228366	doctor :18926	Manager :15747
Mean :231103	master :41950	Training: 8991
3rd Qu.:348236		Director: 7716
Max. :475316		VP : 2776
		(Other) : 4245

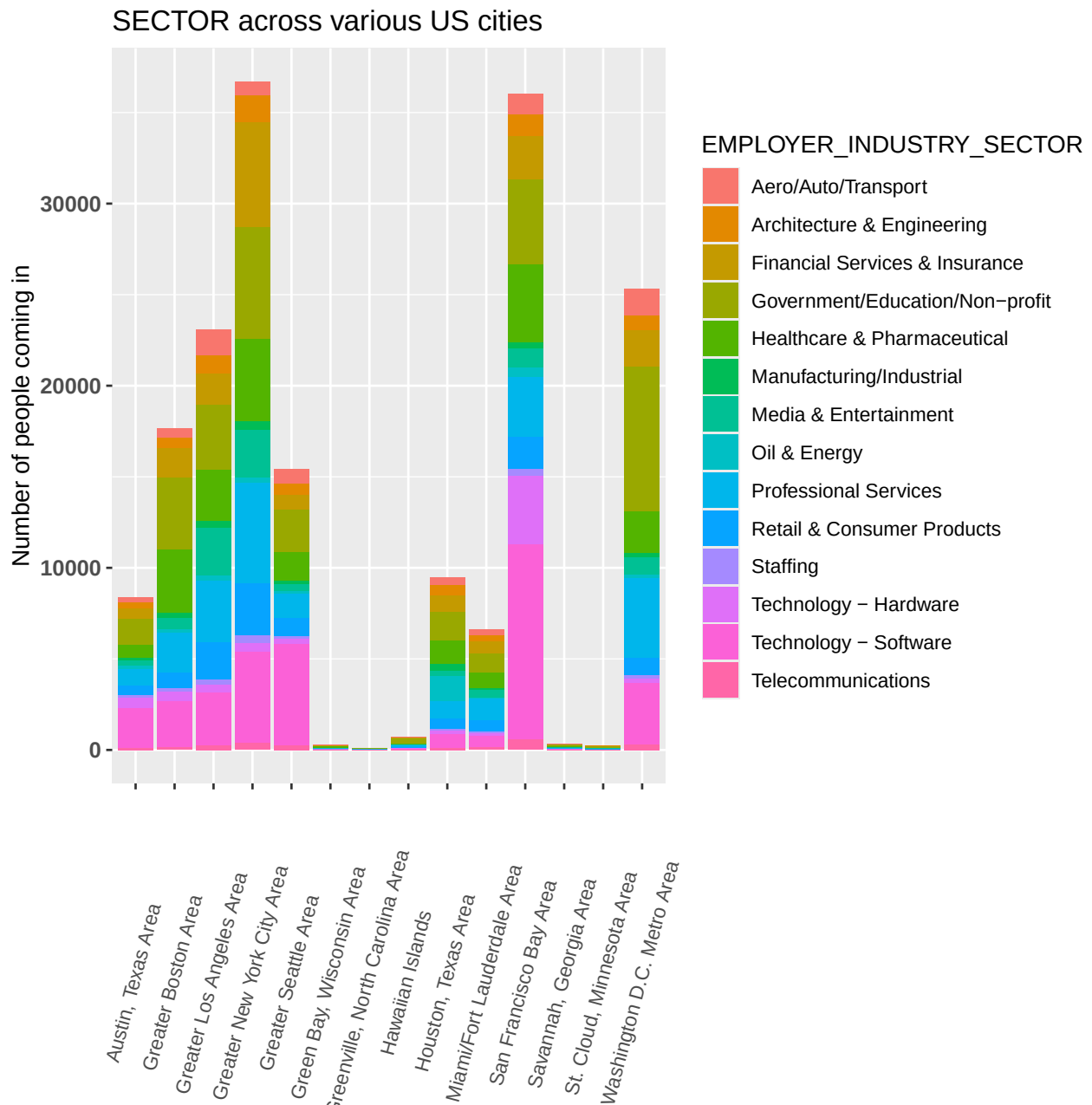
EMPLOYER_INDUSTRY_SECTOR	POSITION_FUNCTION
Technology - Software :33788	Engineering :26427
Government/Education/Non-profit:33090	Education :15830
Professional Services :23385	Research :15050
Healthcare & Pharmaceutical :22185	Sales :12712
Financial Services & Insurance :16524	Business Development:11655
Retail & Consumer Products :11439	Operations :11573
(Other) :40005	(Other) :87169

WEEK_BEGINNING	SOURCE_COUNTRY	SOURCE_REGION
7/31/2016: 4510	Australia : 0	Greater New York City Area: 15572
8/14/2016: 4505	United Kingdom: 0	San Francisco Bay Area : 10268
8/21/2016: 4493	United States :180416	Greater Los Angeles Area : 9648
8/28/2016: 4354		Greater Boston Area : 8611
9/11/2016: 4327		Washington D.C. Metro Area: 7107
6/5/2016 : 4289		Greater Chicago Area : 6984
(Other) :153938		(Other) :122226

DESTINATION_COUNTRY	DESTINATION_REGION
Australia : 0	Greater New York City Area:36695
United Kingdom: 0	San Francisco Bay Area :36040
United States :180416	Washington D.C. Metro Area:25335
	Greater Los Angeles Area :23073
	Greater Boston Area :17690
	Greater Seattle Area :15436
	(Other) :26147

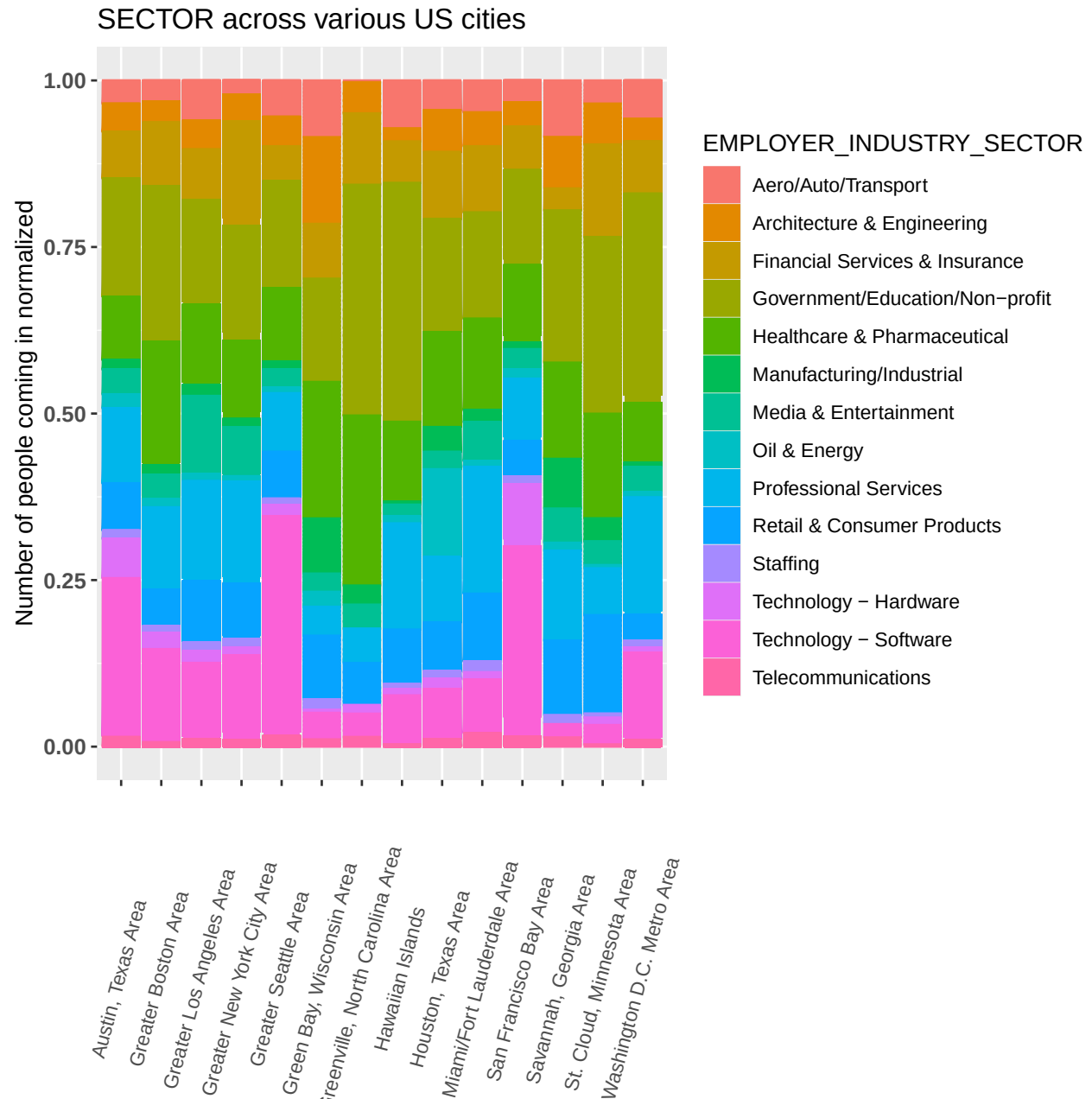
5.1 Bivariate measures of association USA - sector

```
# qplot for visualization
qplot(x = DESTINATION_REGION, data = some_USA, fill = EMPLOYER_INDUSTRY_SECTOR, geom = "bar") +
  theme(axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10, face="bold")) +
  labs(title="SECTOR across various US cities", x="", y="Number of people coming in")
```



```
# ggplot
ggplot(some_USA, aes(x=DESTINATION_REGION, y=NEW_MEM_ID, fill=EMPLOYER_INDUSTRY_SECTOR)) +
  geom_bar(aes(colour = EMPLOYER_INDUSTRY_SECTOR), stat="identity", position = "fill") +
  theme(legend.position = "right", axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10, face="bold")) +
  # scale_fill_brewer(palette = "Dark2") +
  # scale_fill_grey() +
```

```
labs(title="SECTOR across various US cities", x="", y="Number of people coming in normalized")
```



```
# cramer v Sector X CITY OF DESTINATION (top 3)
x<- some_USA$DESTINATION_REGION
y<- some_USA$EMPLOYER_INDUSTY_SECTOR

cv.test = function(x,y) {
  CV = sqrt(chisq.test(x, y, correct=FALSE)$statistic /
    (length(x) * (min(length(unique(x)),length(unique(y))) - 1)))
  print.noquote("Cramér V / Phi:")
  return(as.numeric(CV))
}
```

```
with(some_USA, cv.test(x, y)) # [1] Cramér V / Phi: 0.127 (less )
```

```
## Warning in chisq.test(x, y, correct = FALSE): Chi-squared approximation may be  
## incorrect
```

```
## [1] Cramér V / Phi:
```

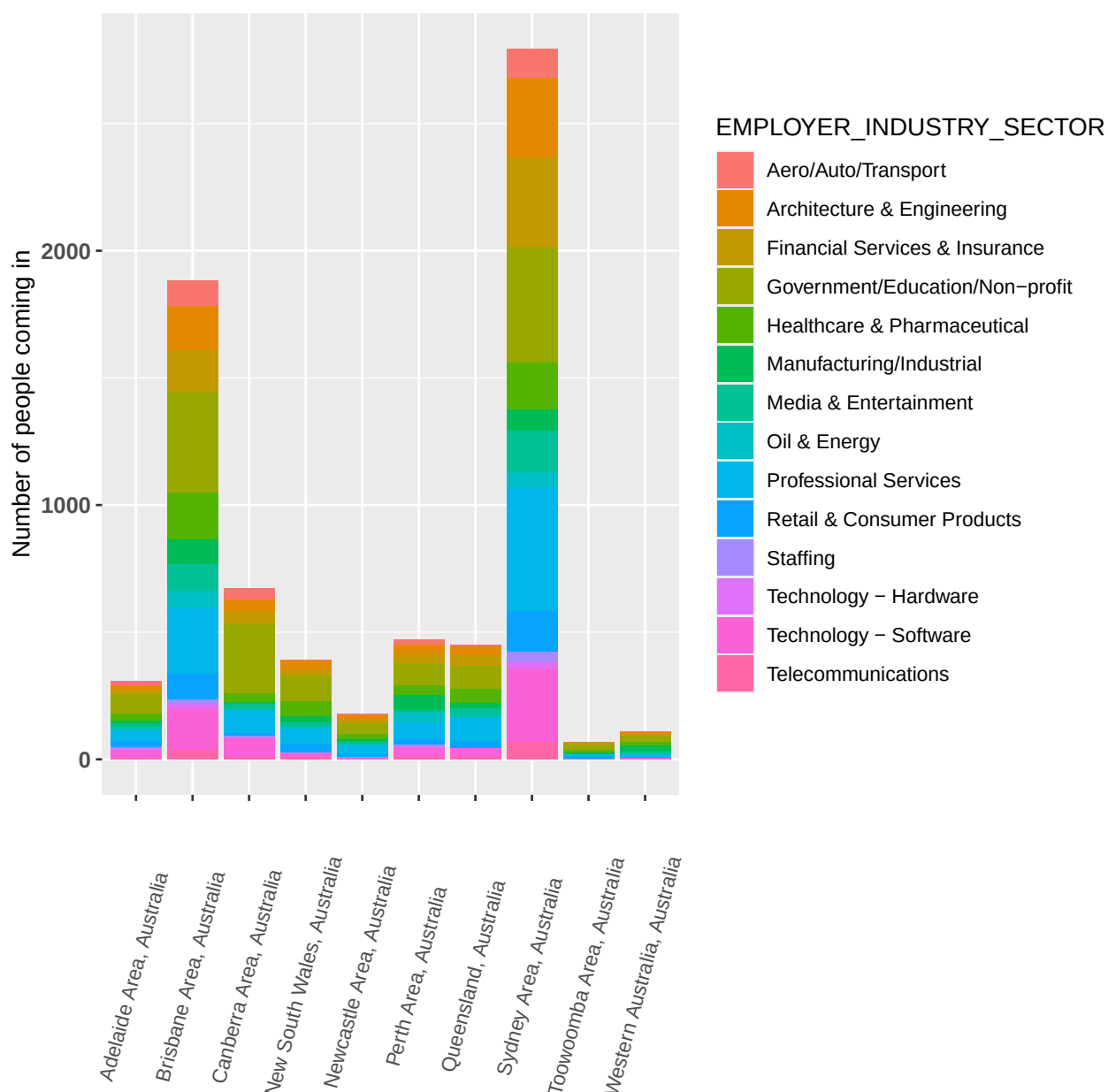
```
## [1] 0.1270301
```

5.2 Bivariate measures of association AUSTRALIA - sector

```
# qplot for visualization
```

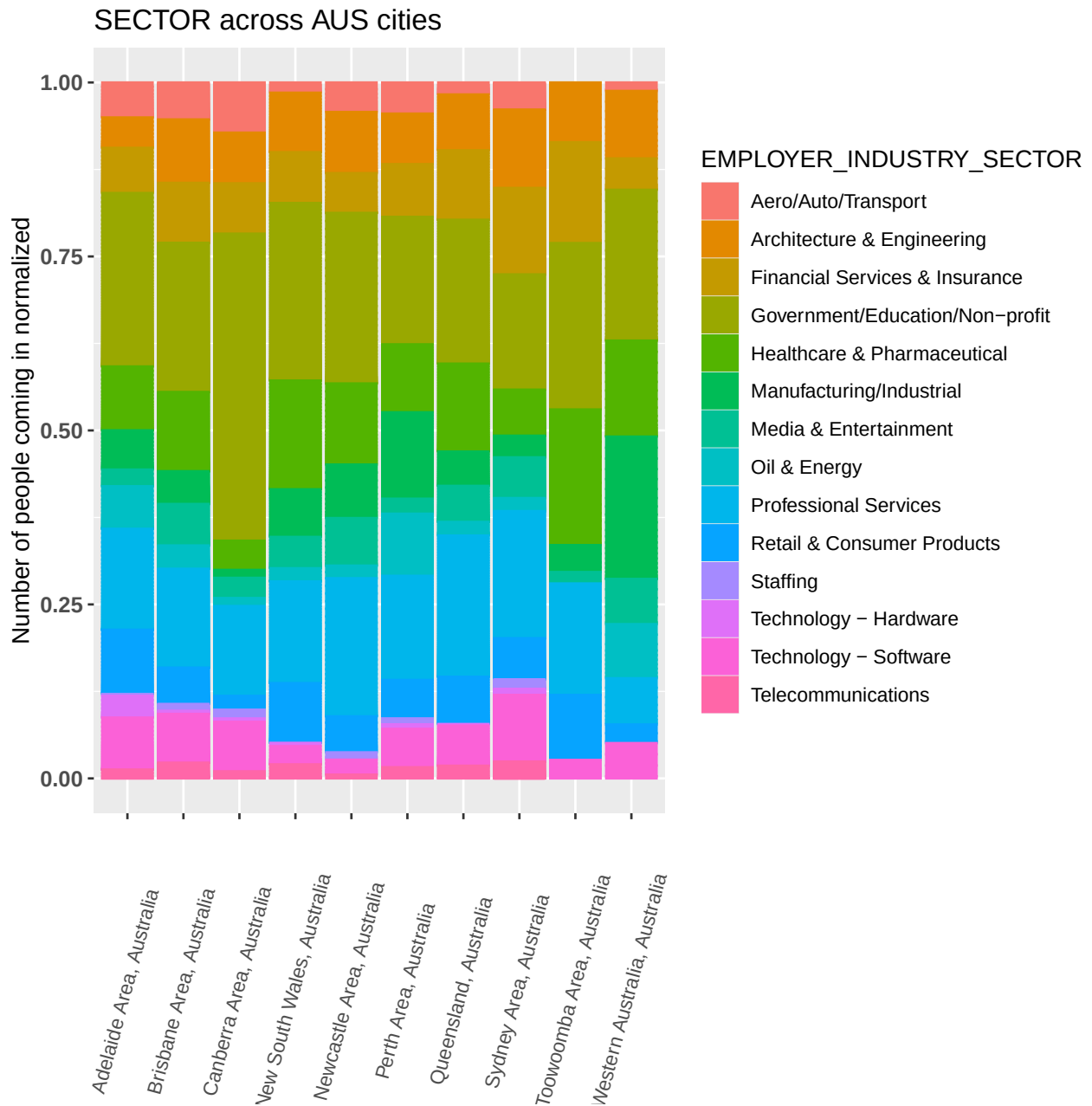
```
qplot(x = DESTINATION_REGION, data = both_AUS, fill = EMPLOYER_INDUSTRY_SECTOR, geom = "bar") +  
theme(axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10,face="bold"))  
labs(title="SECTOR across AUS cities", x="", y="Number of people coming in")
```

SECTOR across AUS cities



```
# ggplot
ggplot(both_AUS, aes(x=DESTINATION_REGION, y=NEW_MEM_ID, fill=EMPLOYER_INDUSTY_SECTOR)) +
  geom_bar(aes(colour =EMPLOYER_INDUSTY_SECTOR),stat="identity", position = "fill") +
  theme(legend.position = "right",axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(angle=0, vjust=0.3))

# scale_fill_brewer(palette = "Dark2") +
# scale_fill_grey() +
labs(title="SECTOR across AUS cities", x="", y="Number of people coming in normalized")
```



```
# cramer v Sector X CITY OF DESTINATION (top 3)
x<- both_AUS$DESTINATION_REGION
y<- both_AUS$EMPLOYER_INDUSTY_SECTOR

cv.test = function(x,y) {
  CV = sqrt(chisq.test(x, y, correct=FALSE)$statistic /
    (length(x) * (min(length(unique(x)),length(unique(y))) - 1)))
  print.noquote("Cramér V / Phi:")
  return(as.numeric(CV))
}

with(both_AUS, cv.test(x, y)) # [1] Cramér V / Phi: 0.127 (less )
```

```
## Warning in chisq.test(x, y, correct = FALSE): Chi-squared approximation may be
```

```
## incorrect
## [1] Cramér V / Phi:
## [1] 0.1074171
```

5.3 Bivariate measures of association UK - sector

```
both_UK_N <- both_UK %>% group_by(DESTINATION_REGION) %>% summarise(numIMM = n())
both_UK_N
```

construct UK internal sub-sample (for simplicity)

```
## # A tibble: 60 x 2
##   DESTINATION_REGION      numIMM
##   <fct>                <int>
## 1 Bath, United Kingdom      160
## 2 Belfast, United Kingdom    66
## 3 Birmingham, United Kingdom 949
## 4 Bournemouth, United Kingdom 76
## 5 Brighton, United Kingdom  245
## 6 Bristol, United Kingdom   754
## 7 Bromley, United Kingdom   138
## 8 Cambridge, United Kingdom  719
## 9 Canterbury, United Kingdom  55
## 10 Cardiff, United Kingdom  226
## # i 50 more rows
```

```
some_UK <- both_UK[both_UK$DESTINATION_REGION == "London, United Kingdom" | both_UK$DESTINATION_REGION == "Bromley, United Kingdom" | both_UK$DESTINATION_REGION == "Cardiff, United Kingdom", ]
```

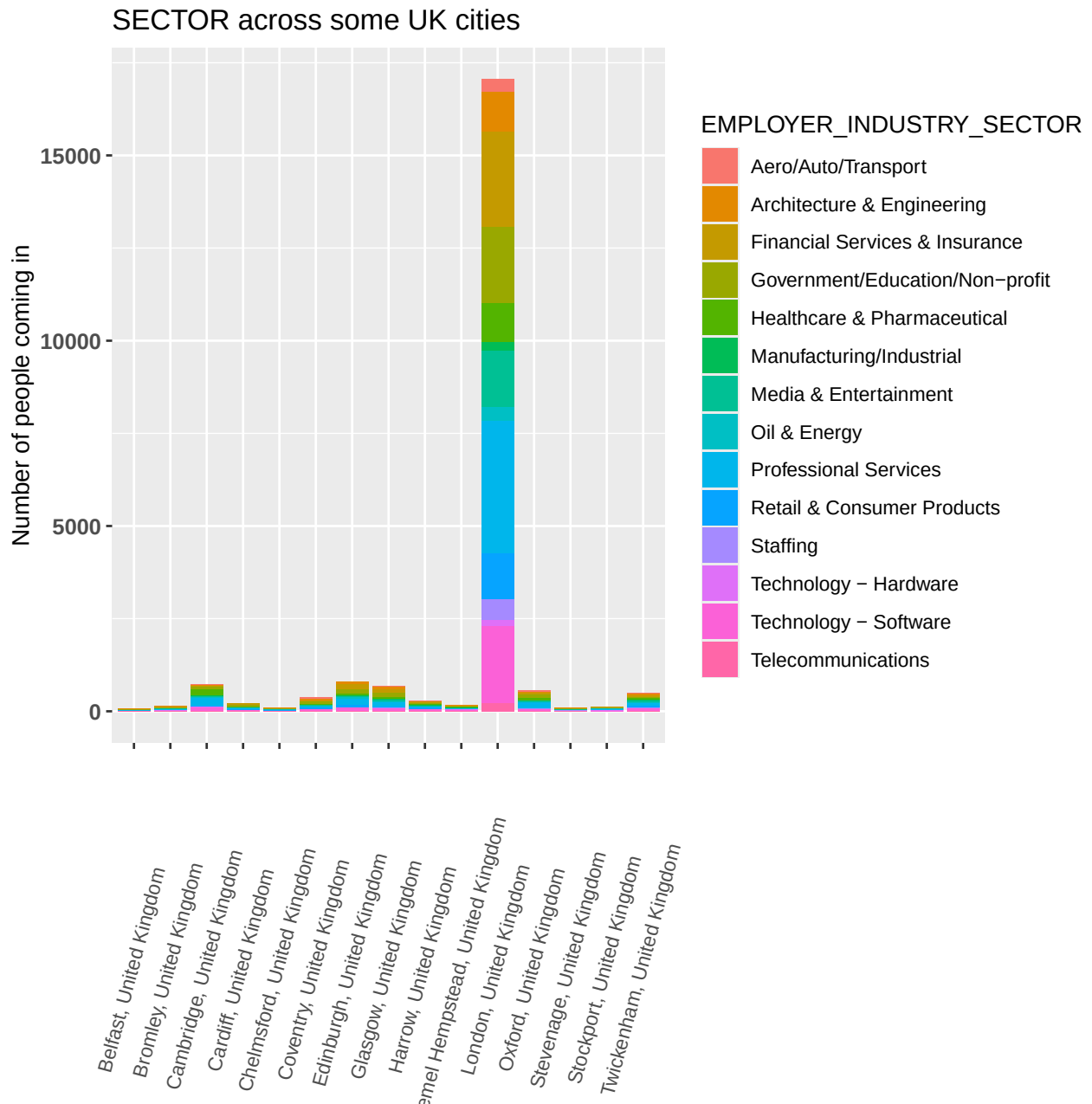
```
summary(some_UK)
```

```
##   NEW_MEM_ID   HIGHEST_DEGREE_OBTAINED   SENIORITY
## Min.      : 18   associate: 677         Entry      :8439
## 1st Qu.:163777   bachelor :13336       Senior      :8351
## Median :286759   doctor   : 2093       Manager     :2520
## Mean    :271096   master   : 5773       Training    :1057
## 3rd Qu.:385509                   Director    : 754
## Max.    :475315                   VP          : 254
##                                     (Other)    : 504
##
##           EMPLOYER_INDUSTRY_SECTOR           POSITION_FUNCTION
## Professional Services      :4481   Business Development: 1927
## Financial Services & Insurance :3093   Sales                : 1773
## Government/Education/Non-profit:2757   Engineering          : 1572
## Technology - Software      :2586   Finance              : 1539
## Media & Entertainment       :1819   Operations           : 1464
## Retail & Consumer Products  :1598   Research             : 1462
## (Other)                    :5545   (Other)              :12142
##
##   WEEK_BEGINNING   SOURCE_COUNTRY
## 10/2/2016: 673   Australia      : 0
## 9/18/2016: 616   United Kingdom:21879
## 9/11/2016: 603   United States  : 0
## 9/25/2016: 583
## 9/4/2016 : 575
```

```
## 10/9/2016: 565
## (Other) :18264
##
## SOURCE_REGION DESTINATION_COUNTRY
## London, United Kingdom : 2192 Australia : 0
## Manchester, United Kingdom : 816 United Kingdom:21879
## Oxford, United Kingdom : 767 United States : 0
## Reading, United Kingdom : 719
## Twickenham, United Kingdom : 712
## Kingston upon Thames, United Kingdom: 699
## (Other) :15974
##
## DESTINATION_REGION
## London, United Kingdom :17055
## Edinburgh, United Kingdom : 803
## Cambridge, United Kingdom : 719
## Glasgow, United Kingdom : 682
## Oxford, United Kingdom : 554
## Twickenham, United Kingdom: 492
## (Other) : 1574
```

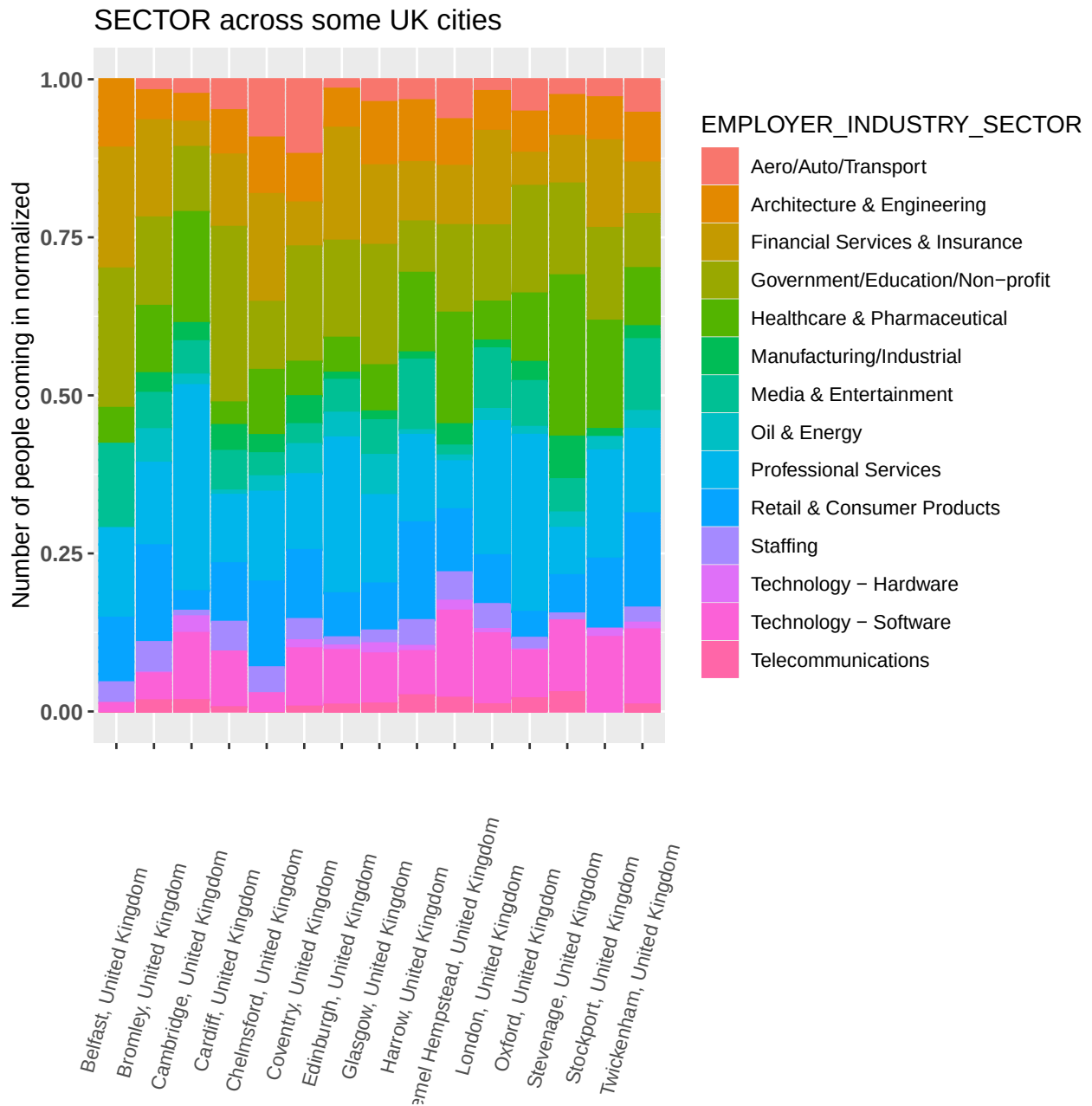
```
# qplot for visualization
```

```
qplot(x = DESTINATION_REGION, data = some_UK, fill = EMPLOYER_INDUSTRY_SECTOR, geom = "bar") +
theme(axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(size=10,face="bold"))
labs(title="SECTOR across some UK cities", x="", y="Number of people coming in")
```



```
# ggplot
ggplot(some_UK, aes(x=DESTINATION_REGION, y=NEW_MEM_ID, fill=EMPLOYER_INDUSTY_SECTOR)) +
  geom_bar(aes(colour =EMPLOYER_INDUSTY_SECTOR),stat="identity", position = "fill") +
  theme(legend.position = "right",axis.text.x = element_text(angle=75, vjust=0.3), axis.text.y = element_text(angle=0, vjust=0.5))

# scale_fill_brewer(palette = "Dark2") +
# scale_fill_grey() +
labs(title="SECTOR across some UK cities", x="", y="Number of people coming in normalized")
```



```
# cramer v Sector X CITY OF DESTINATION (top 3)
```

```
x<- some_UK$DESTINATION_REGION
```

```
y<- some_UK$EMPLOYER_INDUSTY_SECTOR
```

```
cv.test = function(x,y) {
  CV = sqrt(chisq.test(x, y, correct=FALSE)$statistic /
    (length(x) * (min(length(unique(x)),length(unique(y))) - 1)))
  print.noquote("Cramér V / Phi:")
  return(as.numeric(CV))
}
```

```
with(some_UK, cv.test(x, y)) # [1] Cramér V / Phi: 0.127 (less )
```

```
## Warning in chisq.test(x, y, correct = FALSE): Chi-squared approximation may be
```

```
## incorrect
## [1] Cramér V / Phi:
## [1] 0.07060599
```